RainDrop Gaming

Gaming Keypad Assembly Instructions

Revision 0



Overview

- Link to all design files: https://github.com/AGorshkov101/RainDrop-Gaming-Keypad
- See the Bill of Materials (BOM) for all components that need to be ordered or 3D printed
    - I highly recommend ordering a few extras of the PCBA components
- Parts were design for SLA 3D printing. The design is not recommended for FDM

Table of Contents

1. Tools and Skills
2. Flex PCBA
3. Thumb Cluster PCBA
4. Micro PCBA
5. Arduino Sketch Upload
6. PC Software Setup
7. Key Mount Housing Sub Assembly
8. Outer Main Housing Sub Assembly
9. Housing Base Sub Assembly
10. Top Level Assembly
11. KeyLink Tutorial



1

Tools and Skills



Tools

- Soldering Iron
  - Before starting make sure you are familiar with soldering.
  - Tutorial:
    - https://www.youtube.com/watch?v=3jAw41LRBxU
    - https://www.youtube.com/watch?v=p_UKL0Sjy84
- Something to clean soldering tip
- Solder
- Electrical tape (Optional but highly recommended)
  - You can use other aids like soldering helping hands instead if you prefer
- 4-40 thread tap
  - Before starting make sure you are familiar with tapping parts.
  - Tutorial: https://www.youtube.com/watch?v=r6Ijd9o-C10
- Tweezers
- Screwdriver with the following drives
  - Torx Plus IP7, T7
  - Hex 5/64
  - Optional tip - Just get an iFixit kit. It will have both and many more
- Silicone matt to protect your table (Optional but highly recommended)

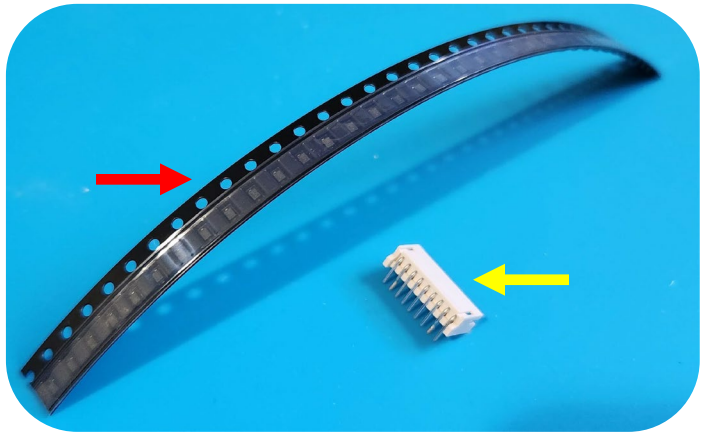


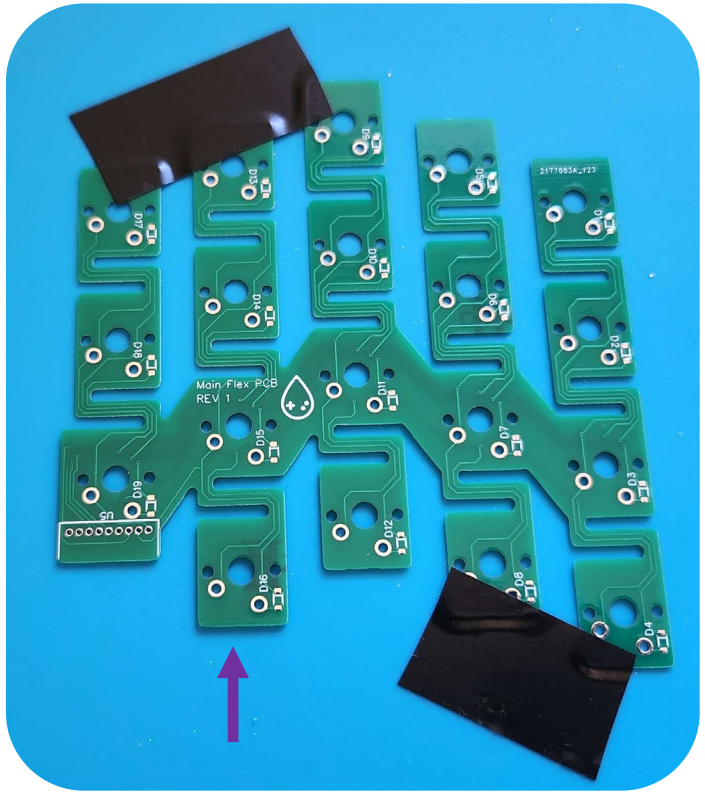
2

Flex PCBA

Components

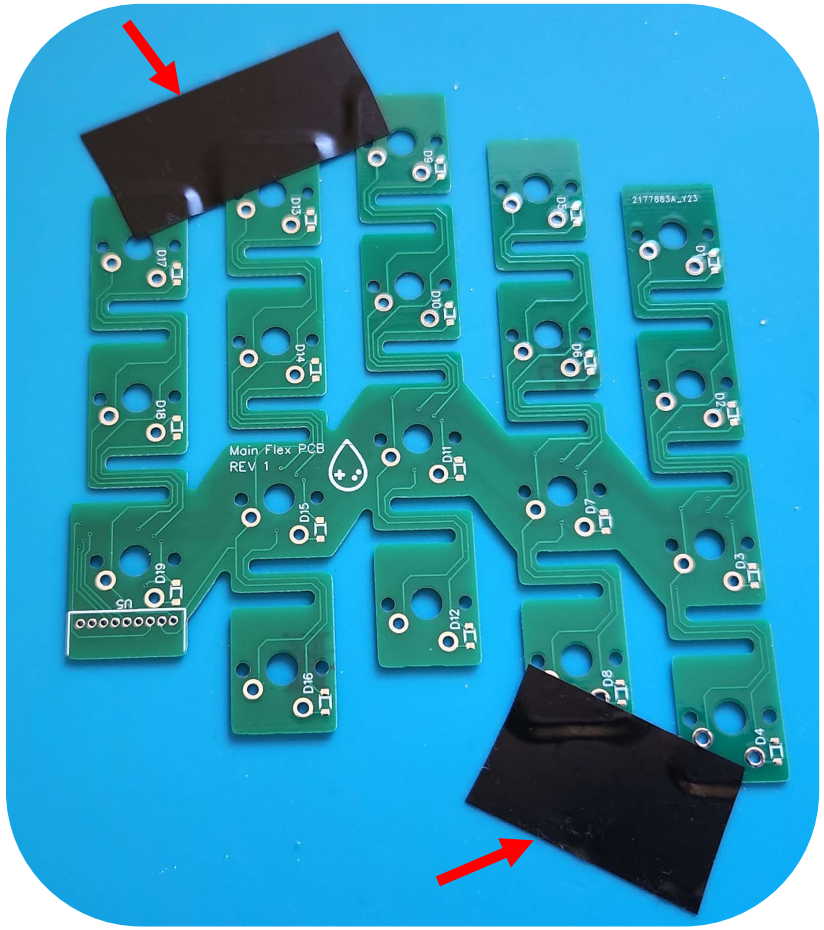
- Diodes
- 9 pin connector
- Flex PCB

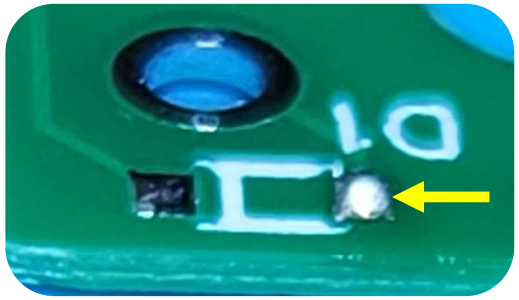


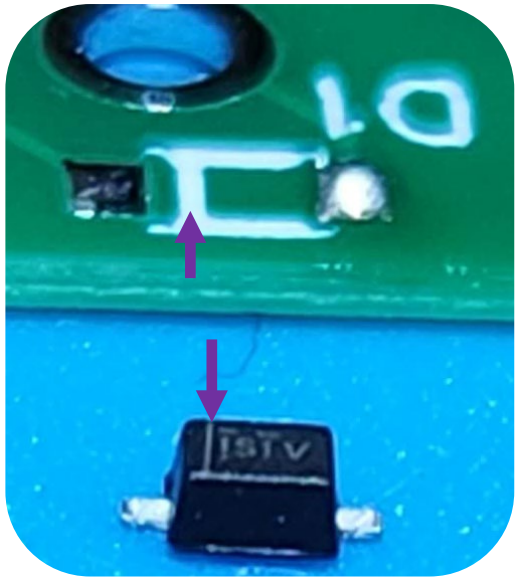


Diode Assembly

- Tape Flex PCB to working surface to stabilize it
- Add solder to one of the pads on the diode footprint
  - Diodes are designated as “D#”, where “#” is an identifying number
- Before soldering the diode to its footprint on the PCB ensure the line indication the orientation is on the same side on both the diode and the footprint
  - You may need to shine a light at an angle on the diodes to make their markings more visible
  - Improper orientation will result in that switch not working

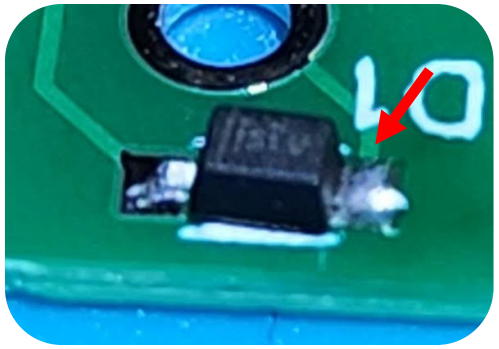




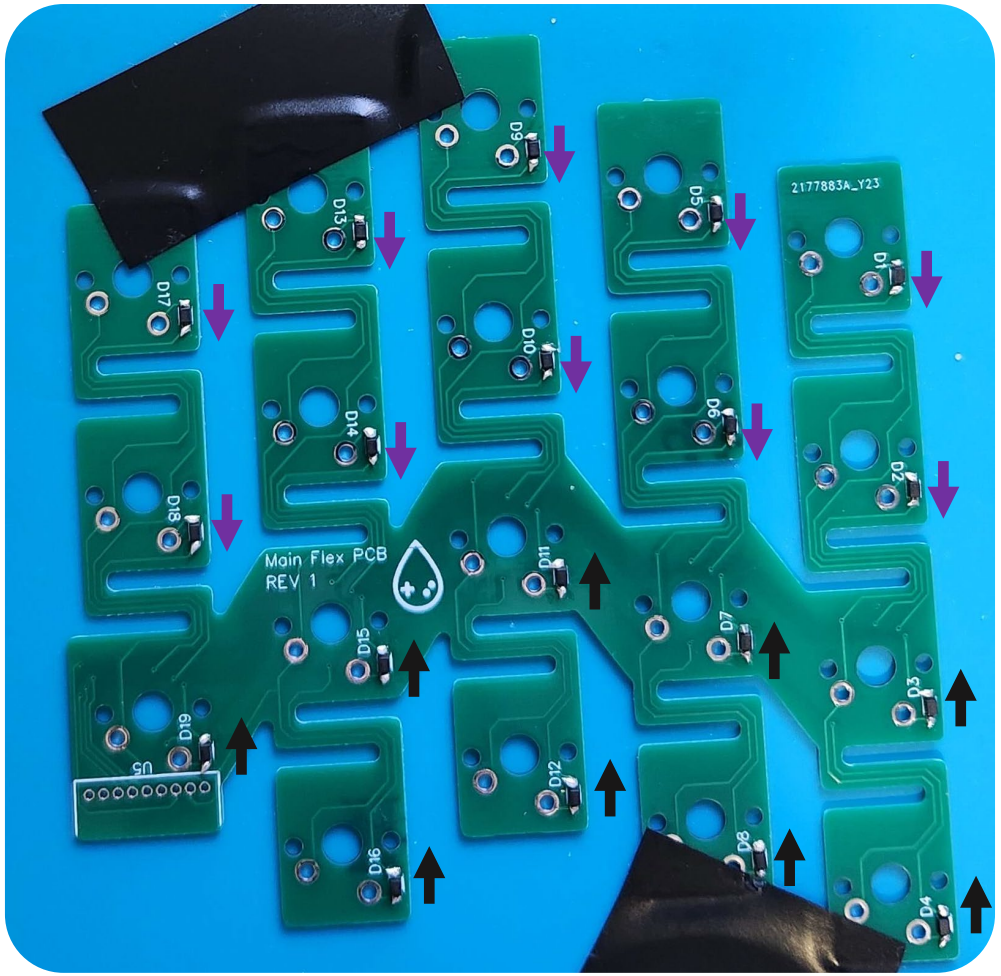


Diode Assembly

- Heat up the pad with solder and place the diode on the foot pad in the correct orientation
- Solder the remaining pad
- Repeat these steps until all diodes are populated
- Some diodes polarity is downward as pictured
- Some diodes polarity is upward as pictured

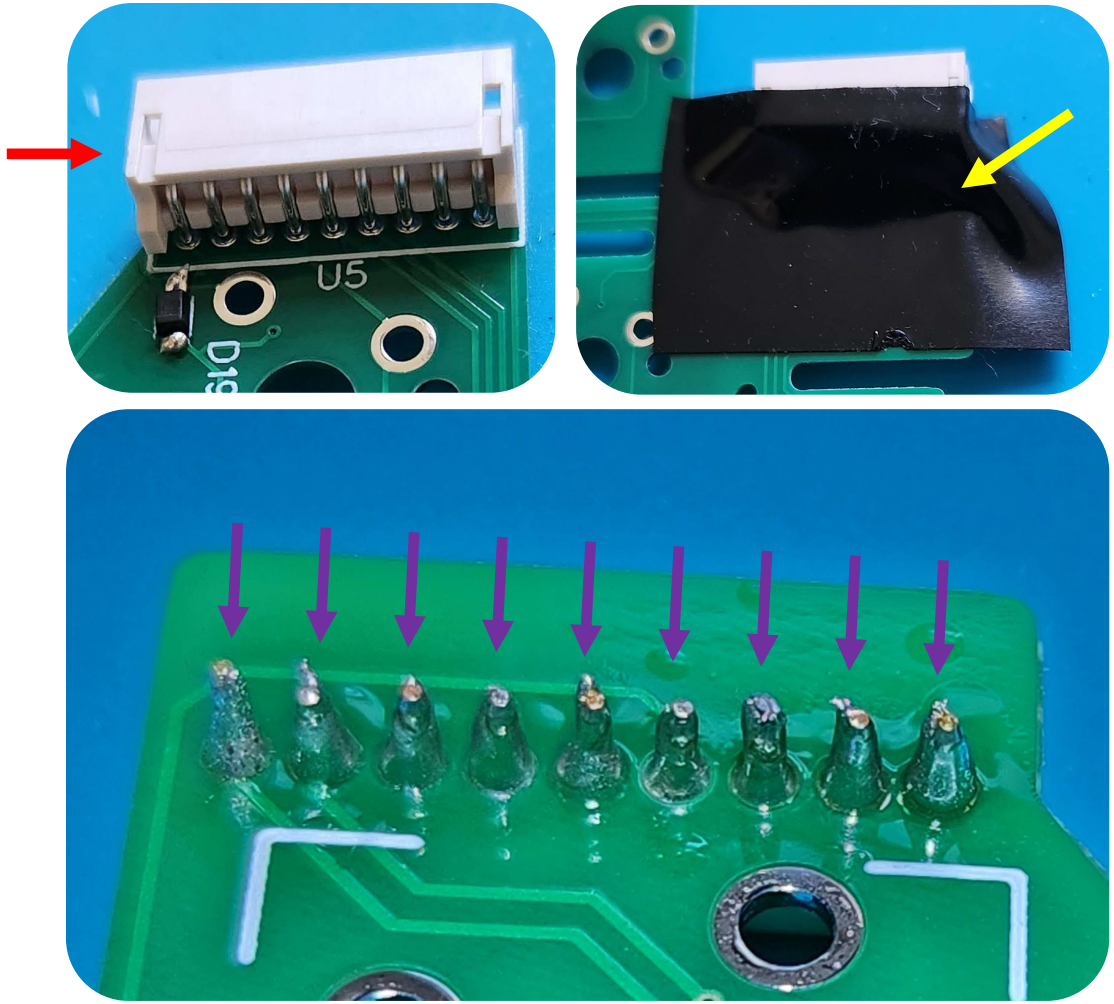






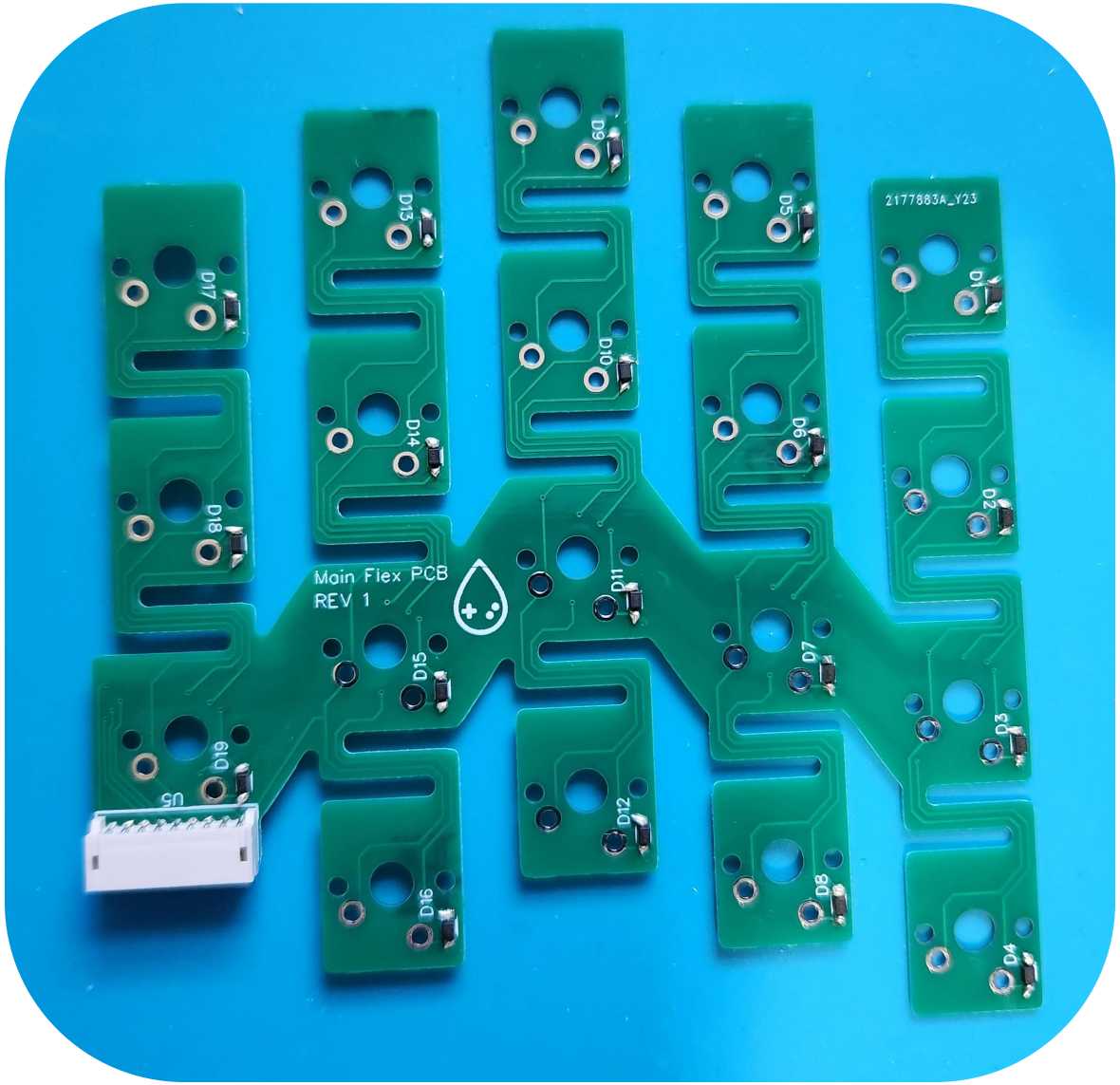
Connector Assembly

- Place the 9-pin connector on its footprint
- Use tape to secure it in place
- Flip it over and solder all the pins
  - Make sure there are no solder bridges connecting two pins together



Completed Assembly

- Completed!
- Don’t worry that was the hardest sub assembly
- Go grab some water…and maybe some chocolate. Treat yourself!
- And yes, I know what you electrical engineers are thinking. I am terrible at soldering…..you are correct.

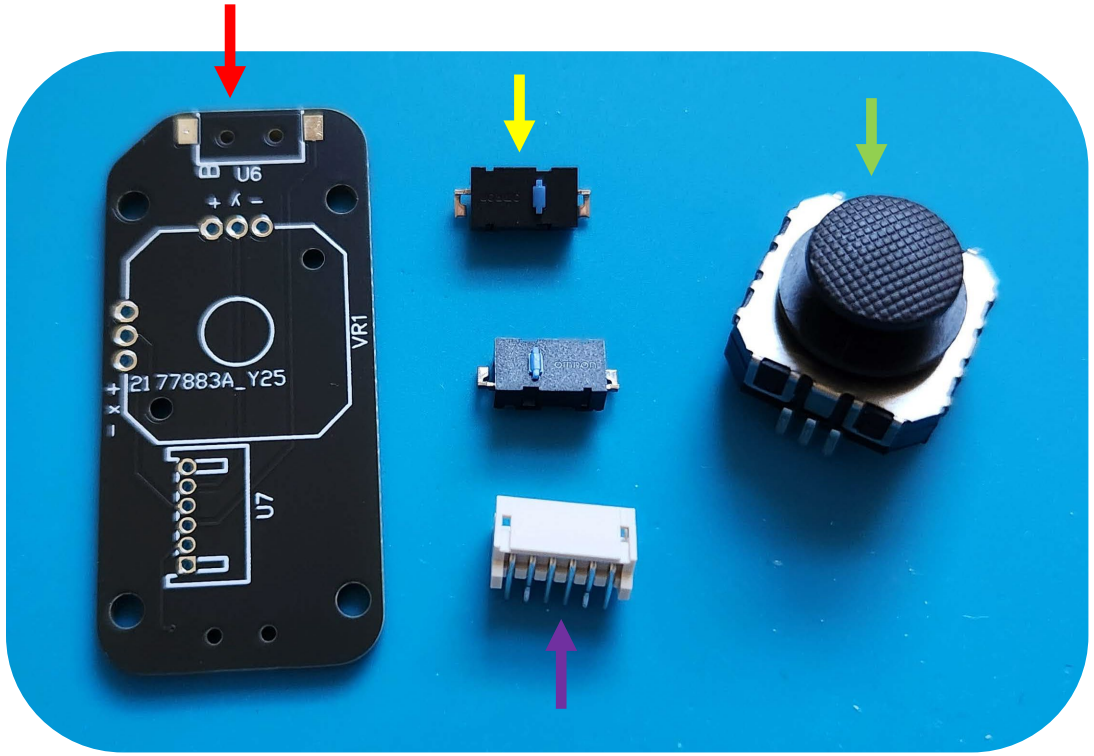


3

Thumb Cluster PCBA

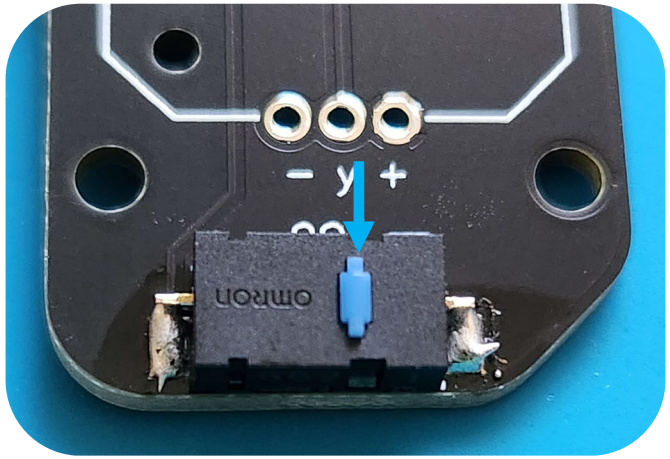
Components

- Thumb cluster PCB
- Omron switch
- 6 pin connector
- Joystick

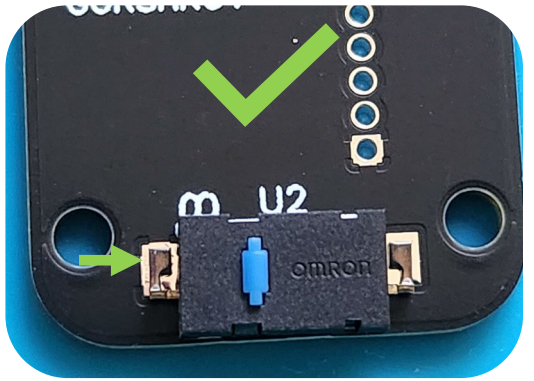


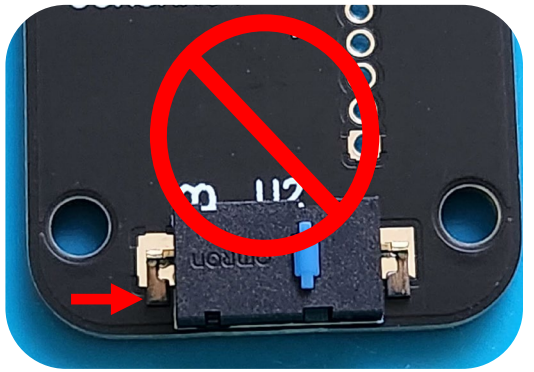
Switch Assembly

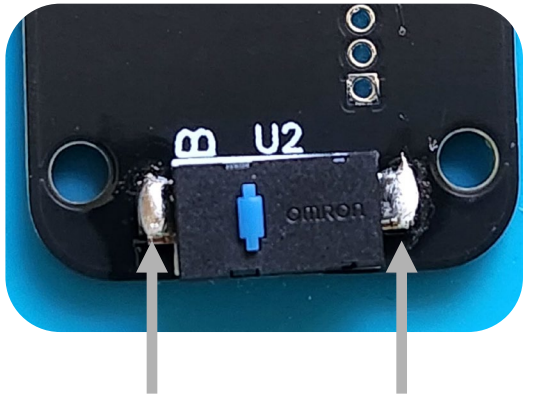
- Flip PCB to the back side and tape to working surface for stability
- Place the Omron switch. The feet should be fully on the solder pads.
- Incorrect orientation
- Solder the Omron switch
- Flip the PCB over and repeat the above steps with the other Omron switch
  - Note that the orientation of the switch is flipped











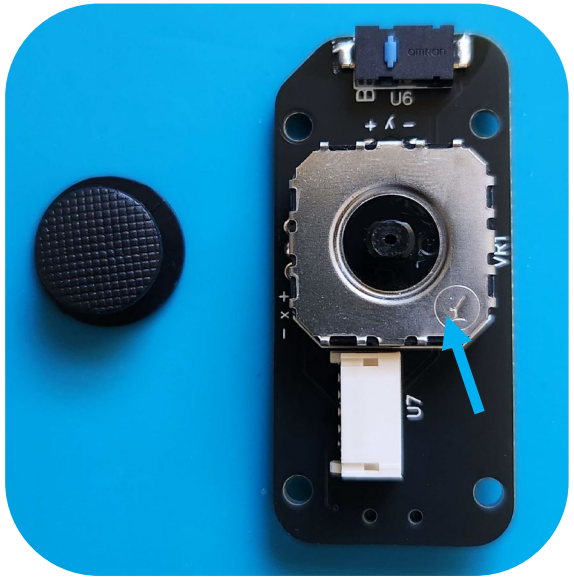
Connector Assembly

- Place 6 pin connector onto the PCB.
  - Pay attention to the orientation
- Secure in place with tape
- Flip over and secure to working surface with tape
- Solder all pins. Ensure not solder bridges exist between adjacent pins



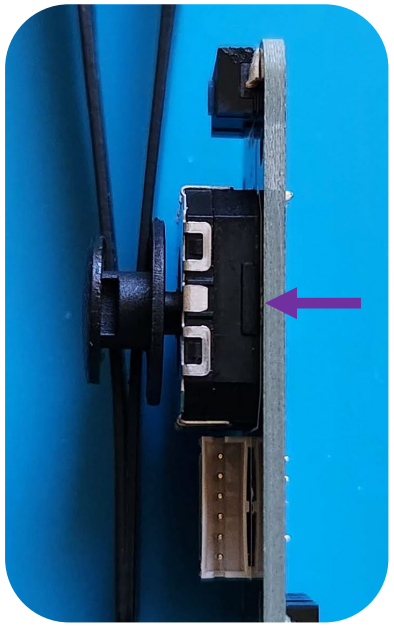
Joystick Assembly

- Place the Joystick. The cap can be removed to make the PCB easier to handle
- Securely tape the Joystick to the PCB so it does not move during soldering
  - It is important that joystick is fully flush against the PCB
- Solder the 6 pins.
  - Ensure there are no solder bridges between the pins
- Check again that the joystick is flush with the PCB on all sides
  - Place the cap back on if removed







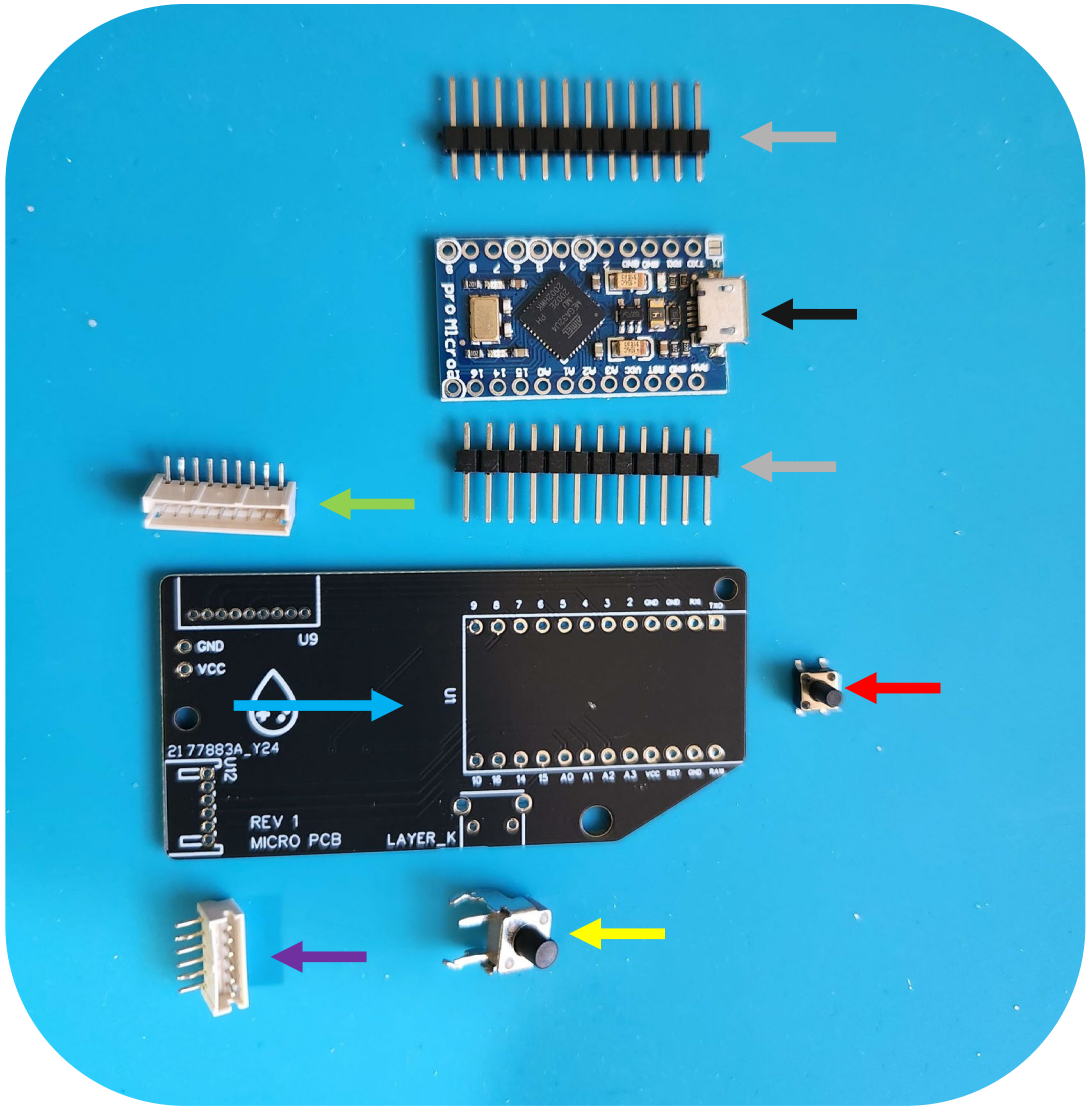


4

Micro PCBA

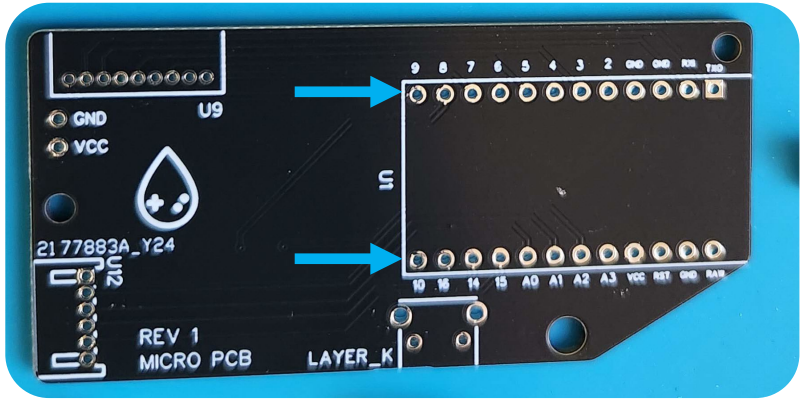
Components

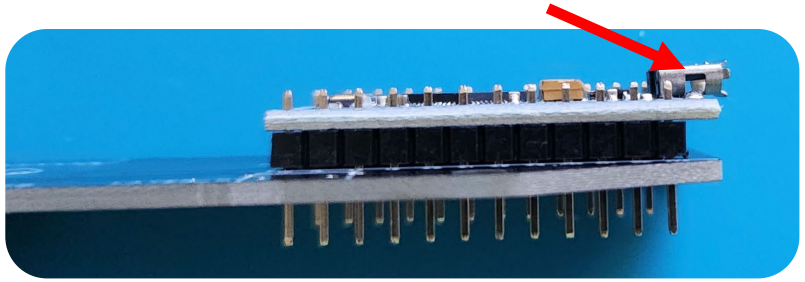
- Micro PCB
- Reset switch (optional)
- Layer button
- 6 pin connector
- 9 pin connector
- Arduino Pro Micro
- Header pins

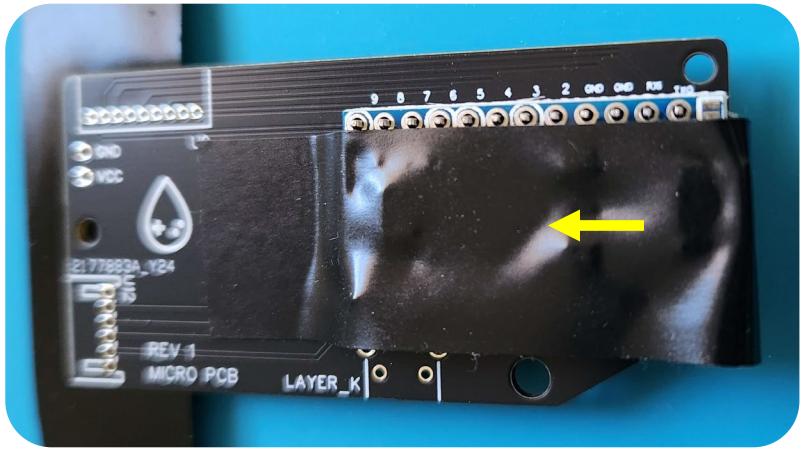


Arduino Assembly

- Place the long pin side of the header pins in the 2 depicted rows of the Micro PCB
- Place the Arduino Pro Micro on top of the short pins of the header pins with the Micro USB connector facing outward
- Secure the 3 component together with some tape.

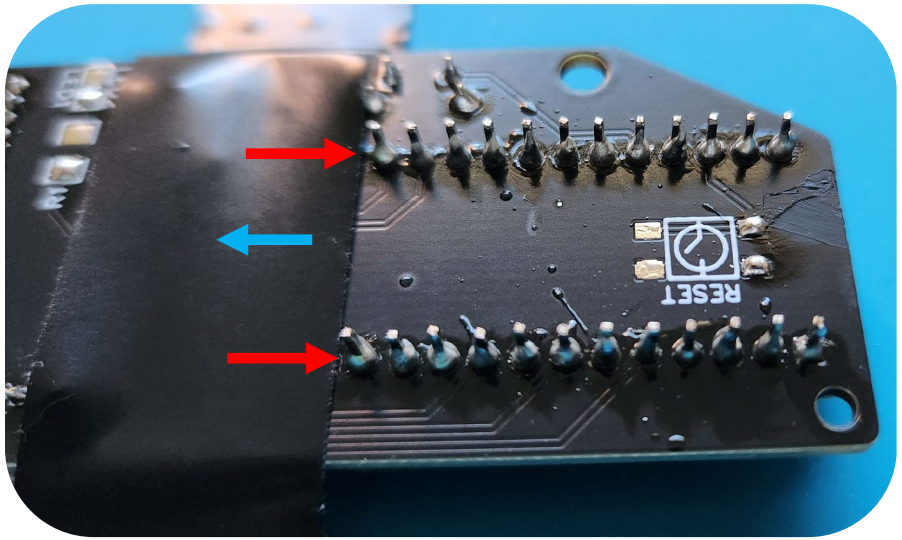


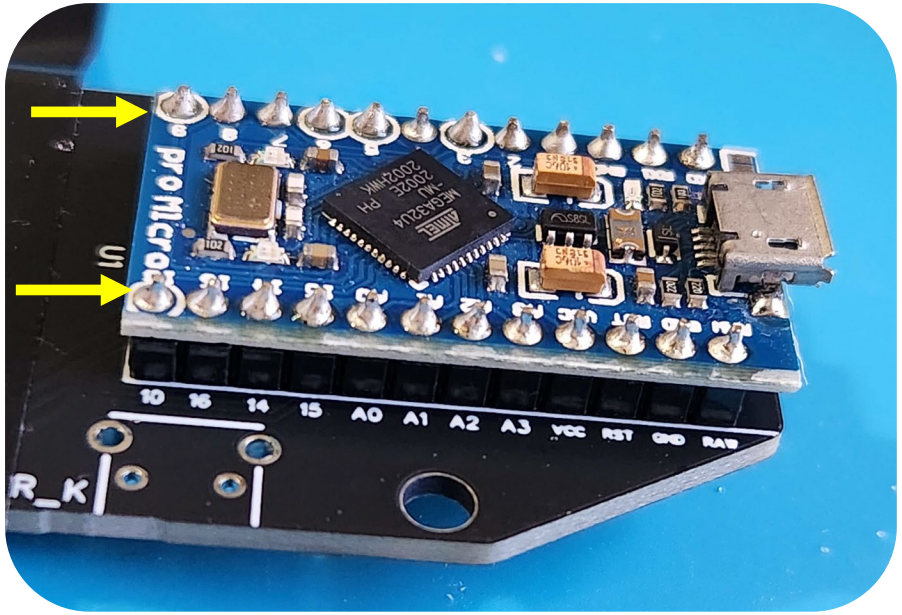




Arduino Soldering

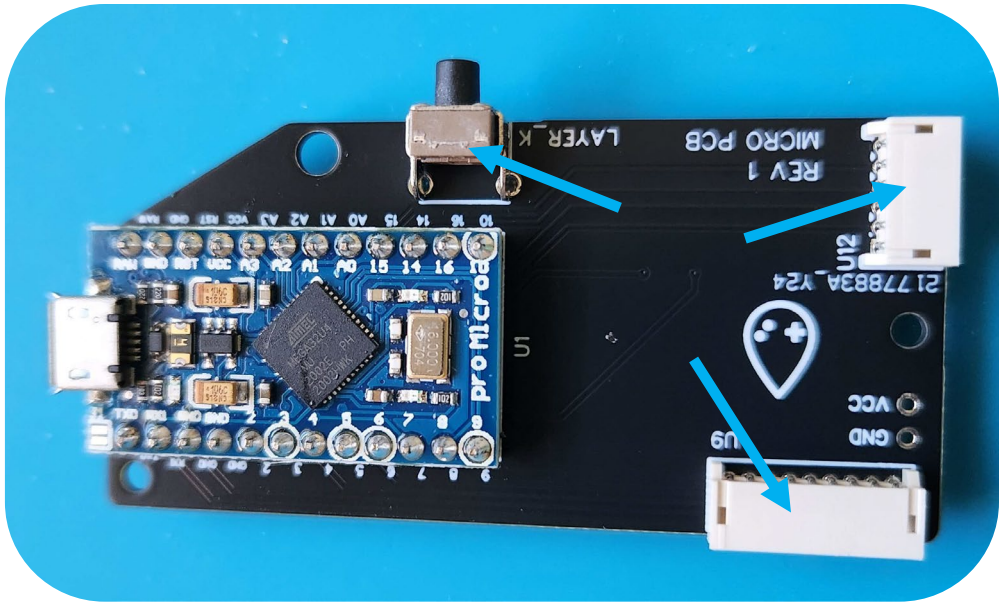
- Flip the assembly upside down and secure it with some tape
- Solder the two rows of header pins to the Micro PCB
- Flip over, secure with tape and solder the two rows of the header pins to the Arduino Pro Micro

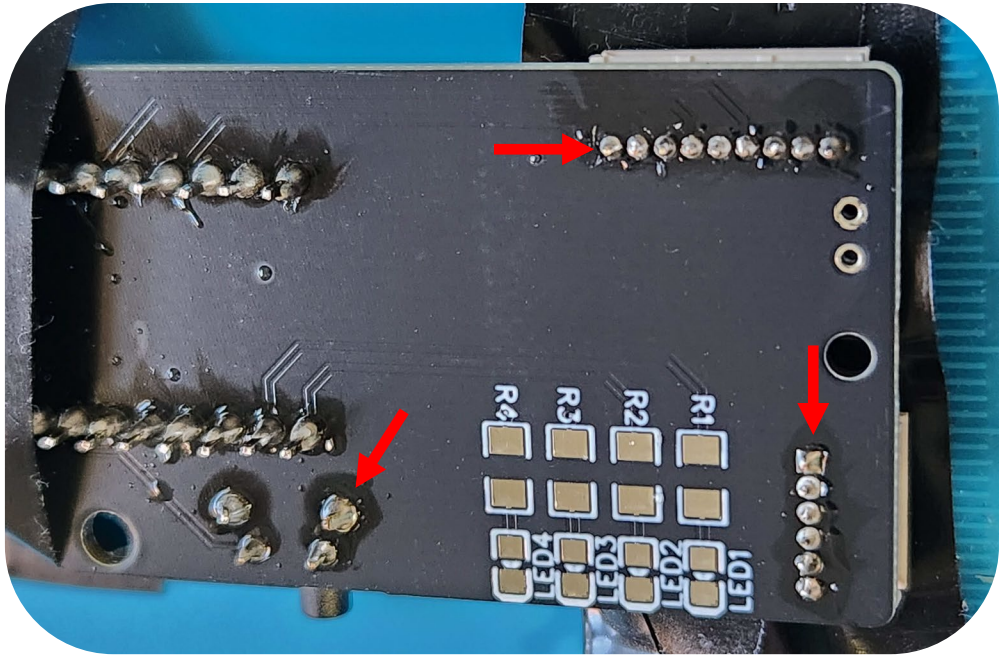




Through Hole Components

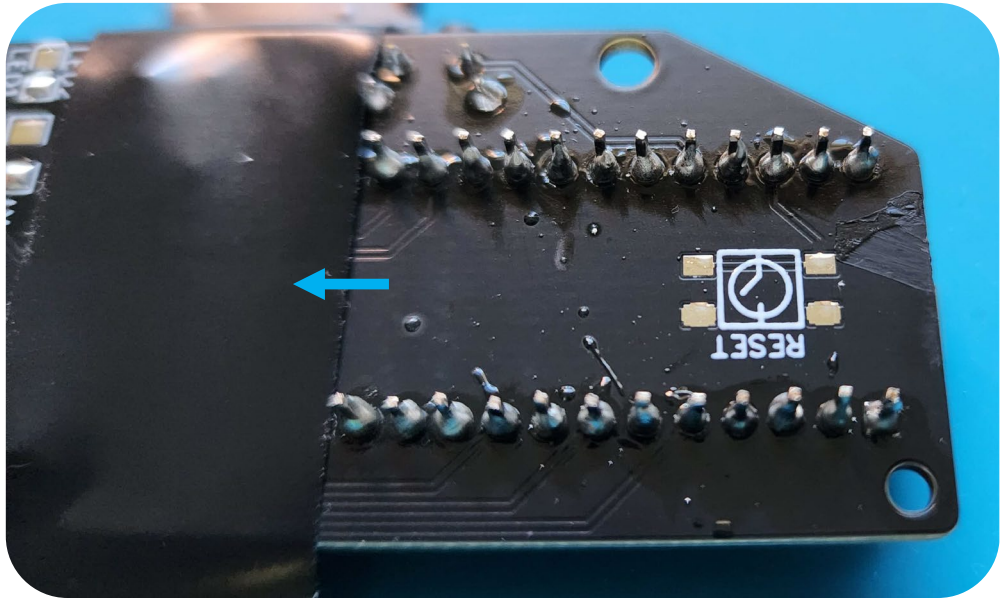
- Place the layer button and two connectors on the Micro PCB
- Flip over, secure assembly with tape, and solder each of their pins

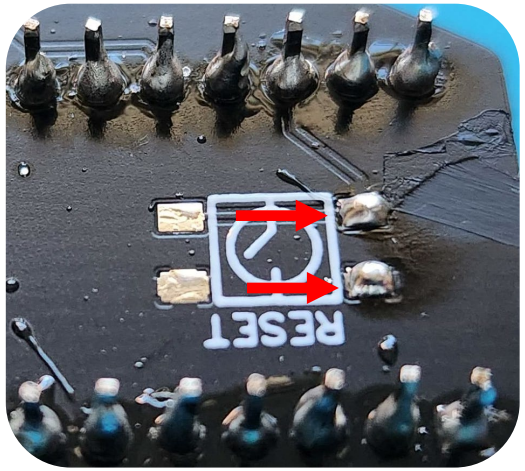


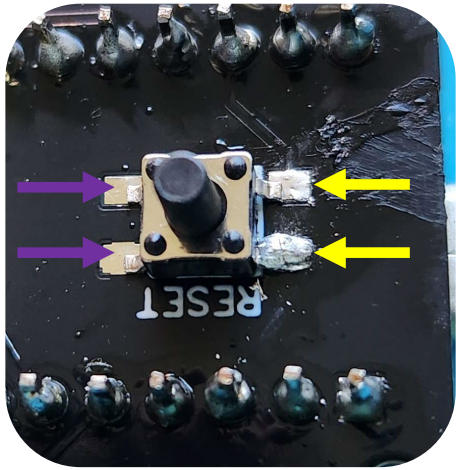


Reset Switch Assembly(Optional)

- Secure PCB with tape
- Add solder to two of the pads
- Heat up the pads and attach the reset switch
- Solder the remaining pads

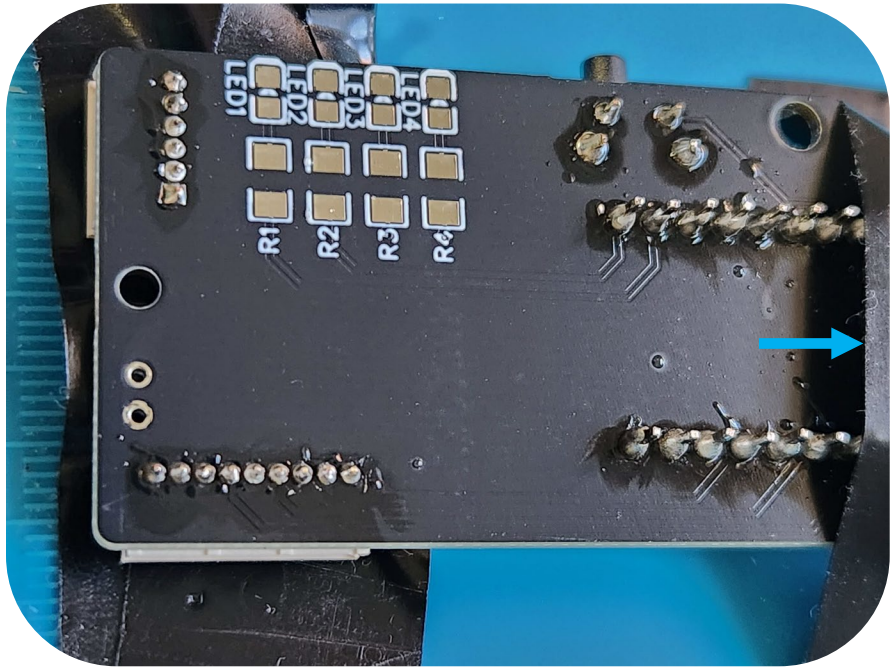


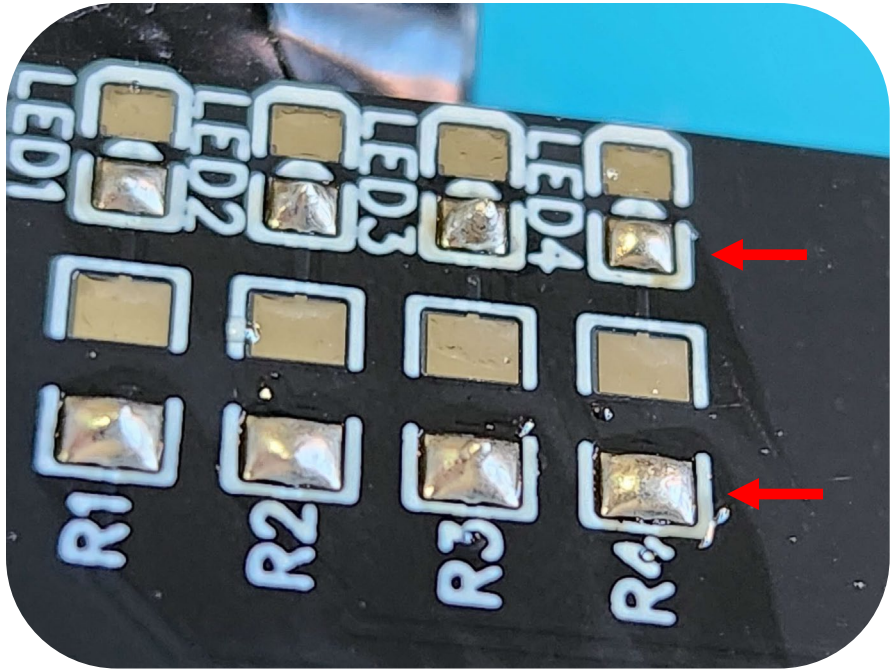




Resistor Assembly

- Secure the assembly with some tape
- Add solder to one of the two feed for the resistors and LED footprints

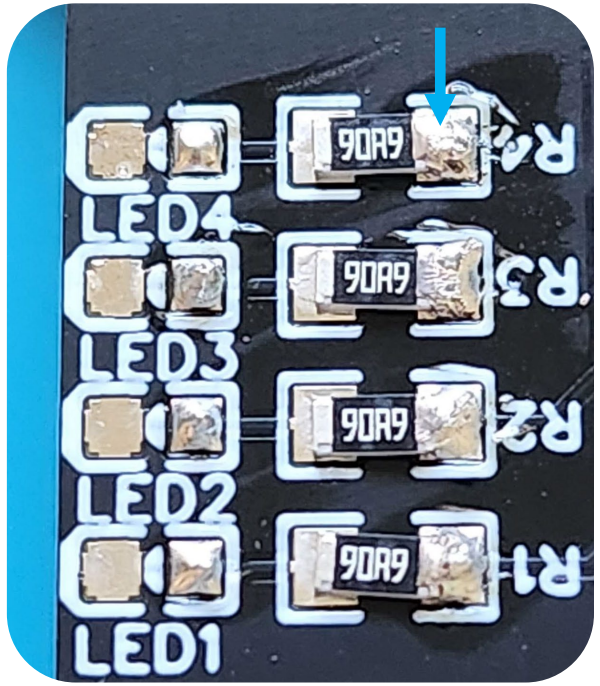


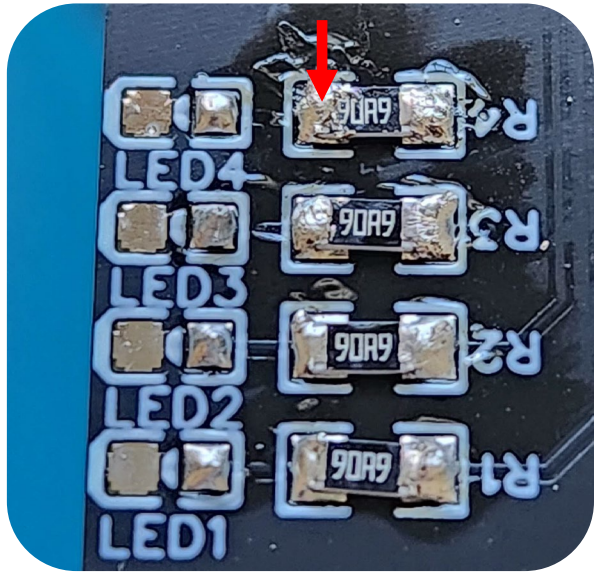


Resistor Assembly

- Heat the soldered pad and insert the resistor
  - Direction dose not matter as long as the side with numbers is up
- Solder each of the remaining foot pads

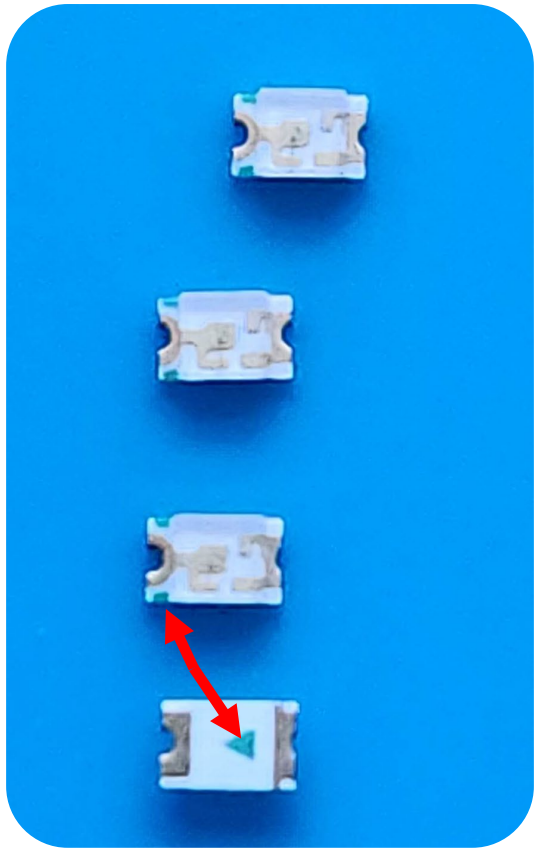


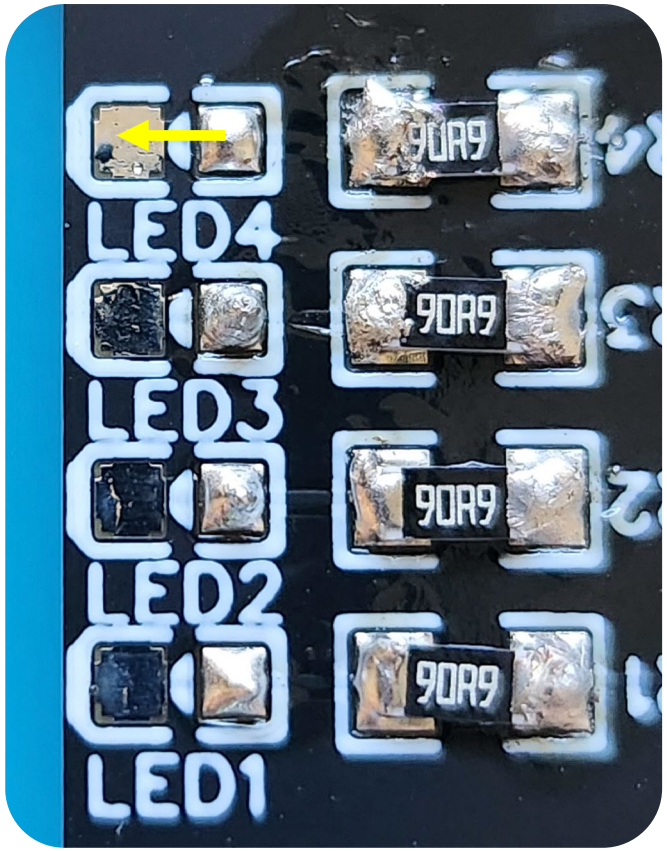


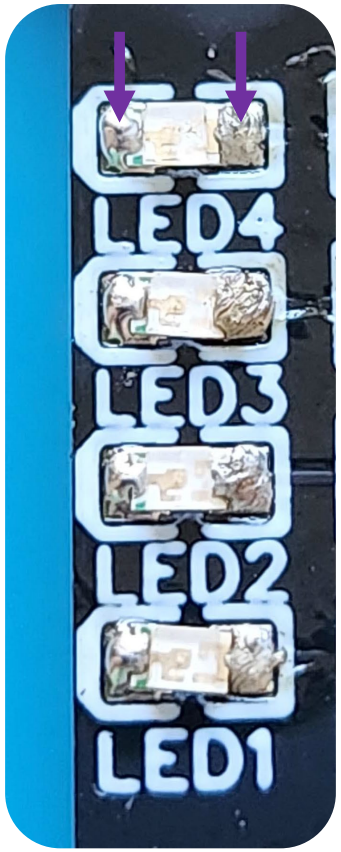


LED Assembly

- Notice how the green arrow on the bottom of the LED points in the direction of the green stripes on the top side
- The arrow from the red step above need to point in the same direction as the arrow on the LED footprint, towards the closest edge of the PCB
  - Double check before soldering or the LED will not work
- Attach the LEDs using the same technique as the resistors

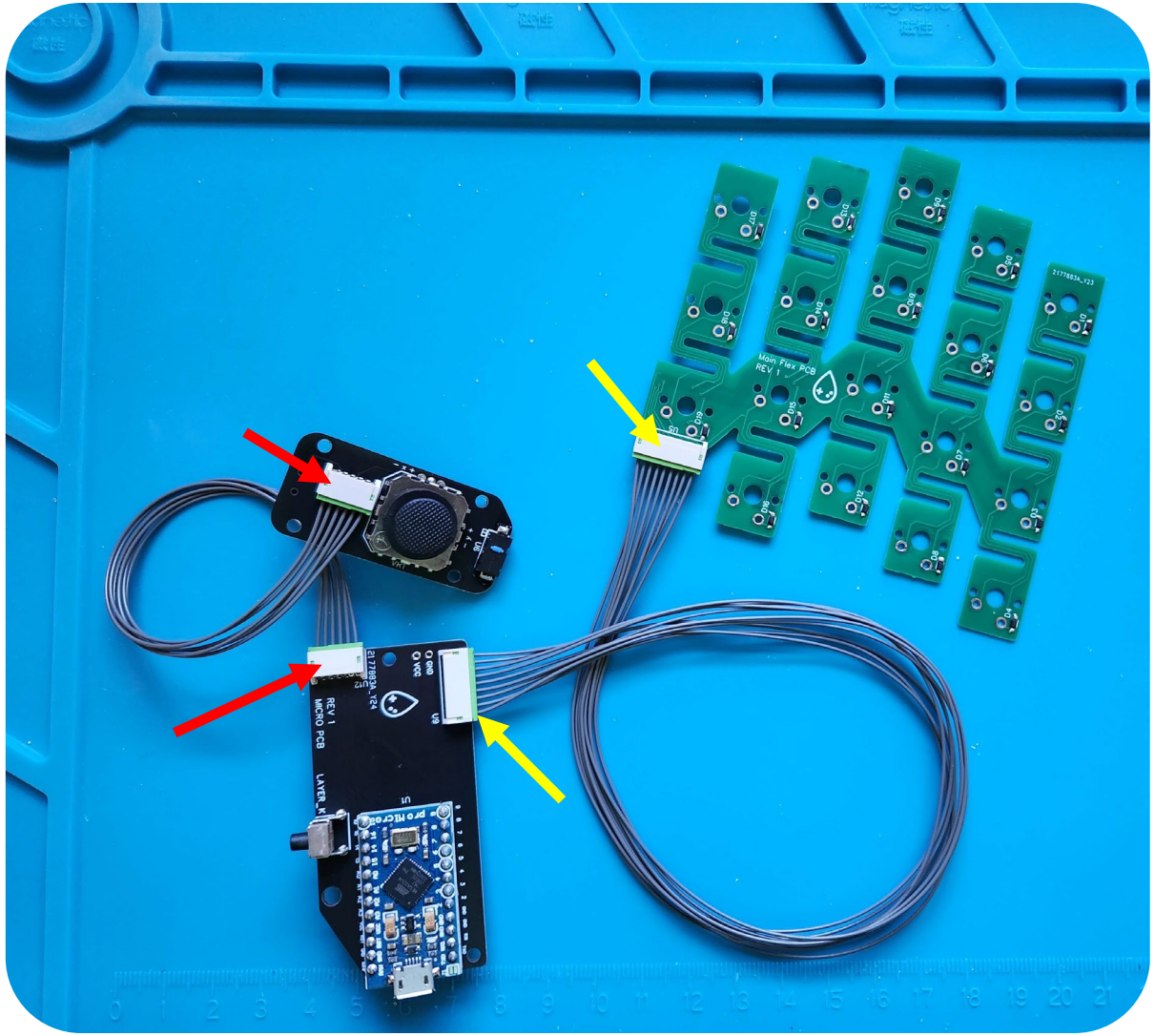






Cable Assembly

- Attach the 6-pin cable to the 6-pin connectors
- Attach the 9-pin cable to the 9-pin connectors



5

Arduino Sketch Upload



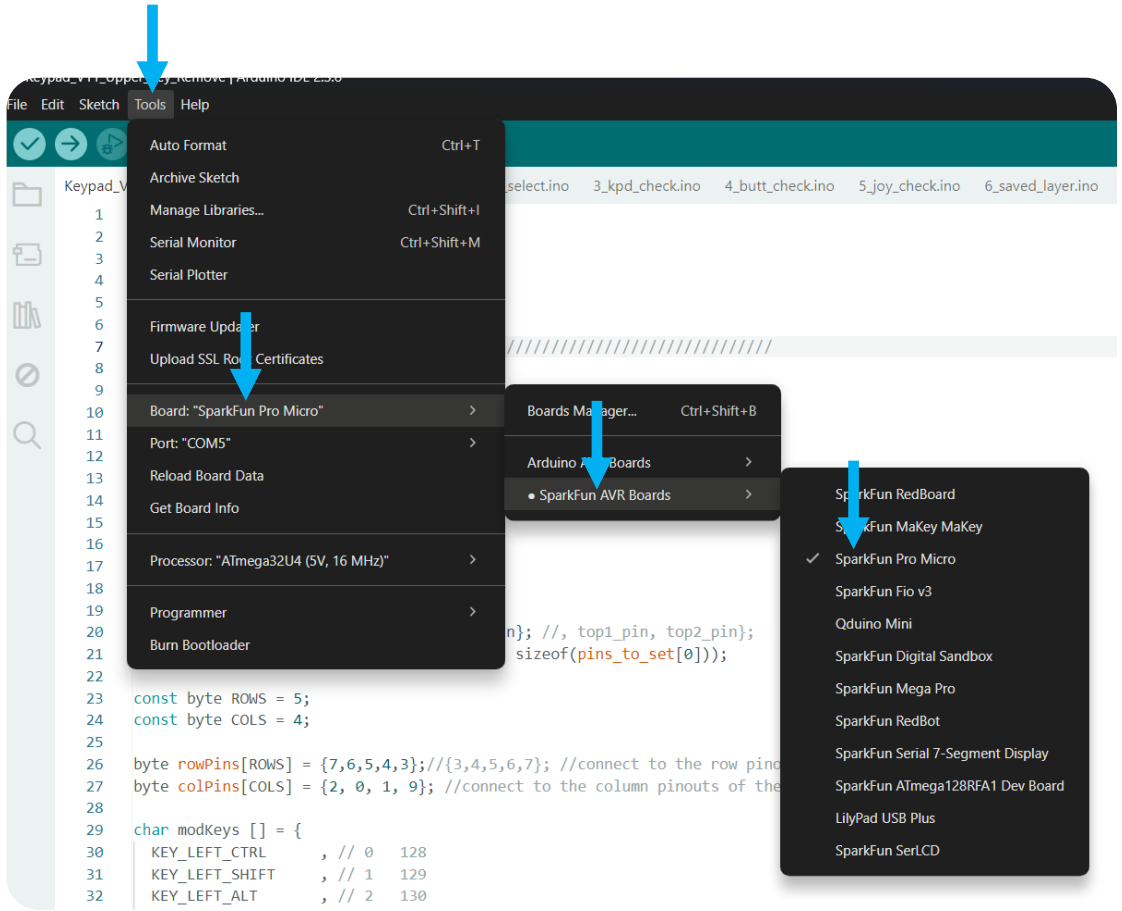
Install Arduino IDE and Board Manager

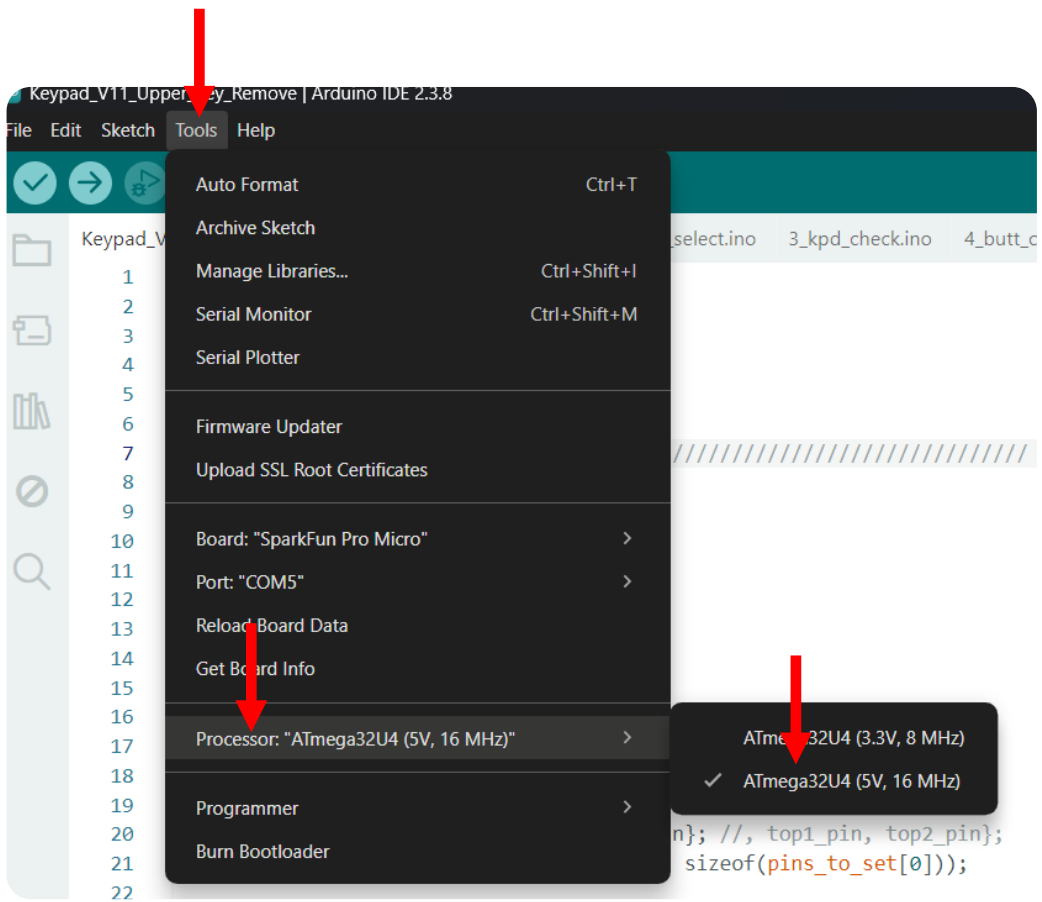
- Download the Arduino IDE and install it
  - https://www.arduino.cc/en/software
- You will need to install the SparkFun board manager library. Follow the steps under “Installing a Third-Party Board Definition” in the guide linked below.
  - https://learn.sparkfun.com/tutorials/installing-board-definitions-in-the-arduino-ide/all#installing-a-third-party-board-definition
  - You only need the following library to be added but feel free to add all 4 mentioned in the guide.
    - https://raw.githubusercontent.com/sparkfun/Arduino_Boards/main/IDE_Board_Manager/package_sparkfun_index.json



Select The Correct Board

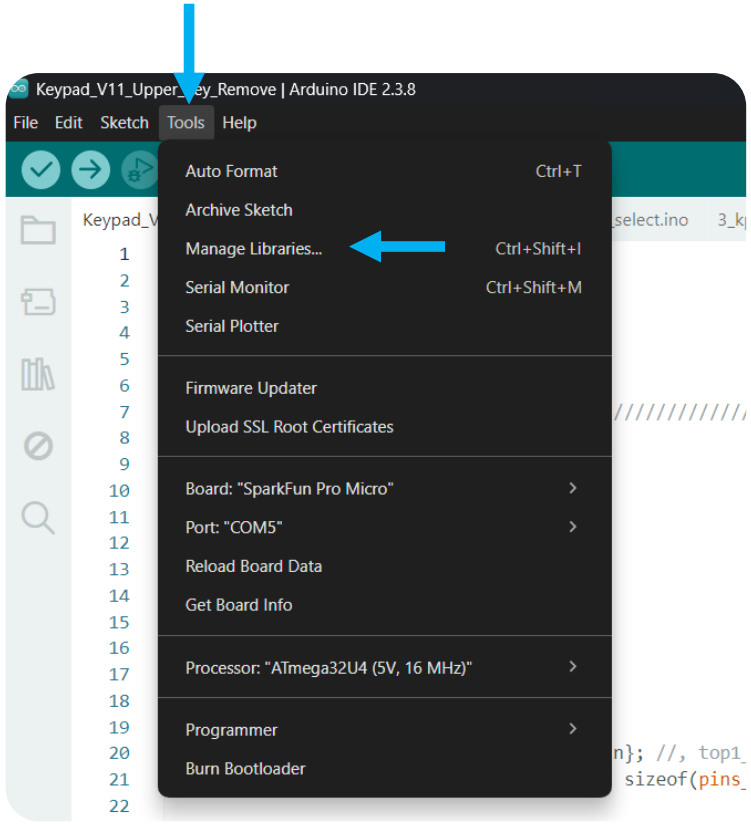
- Select the board: Tools -> Board -> Sparkfun AVR Boards -> Sparkfun Pro Micro
- Select the processor: Tools -> Processor -> 5V

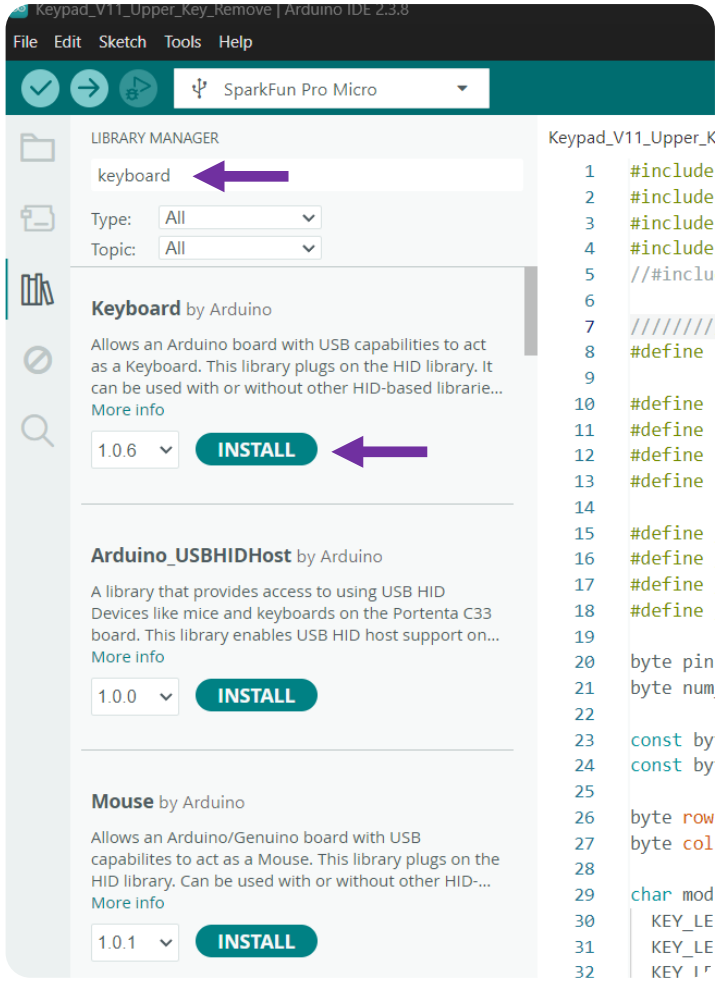




Adding Libraries

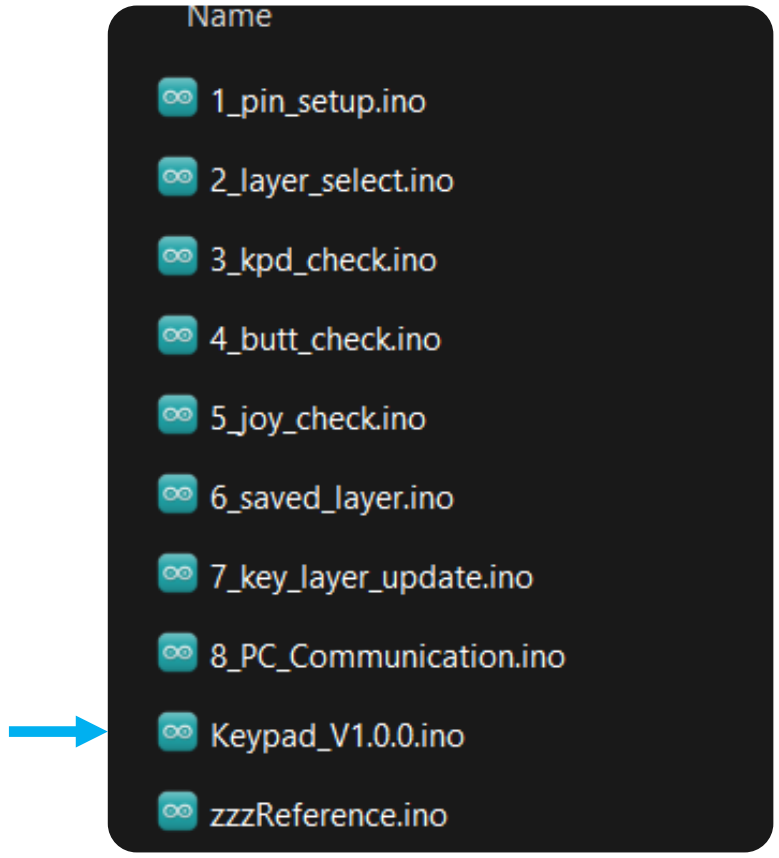
- Start the Arduino IDE
  - Go to tools -> Manage Libraries
- For each of the listed libraries below, type the library name in the search and install
  - Keyboard
  - Keypad

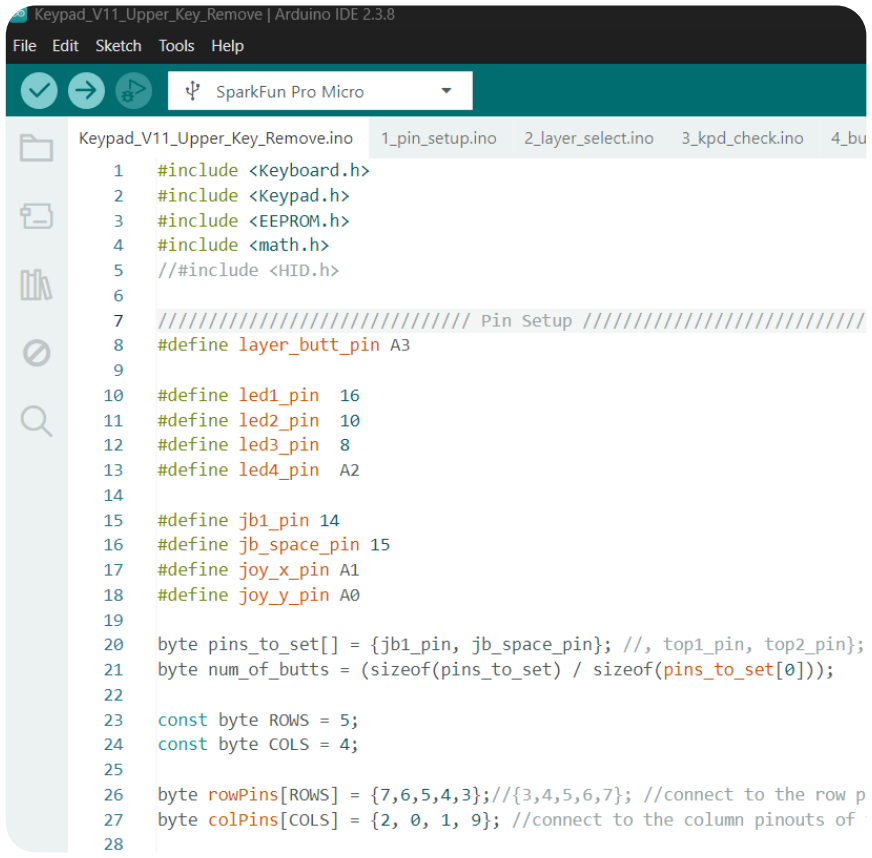




Load The Sketch

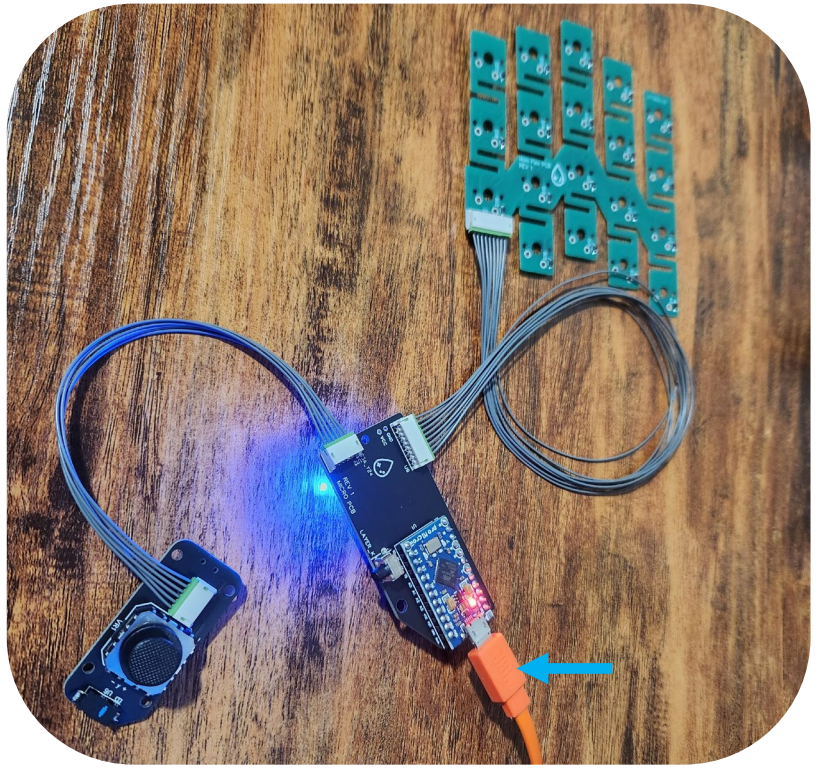
- In file explorer open the Keypad.ino file

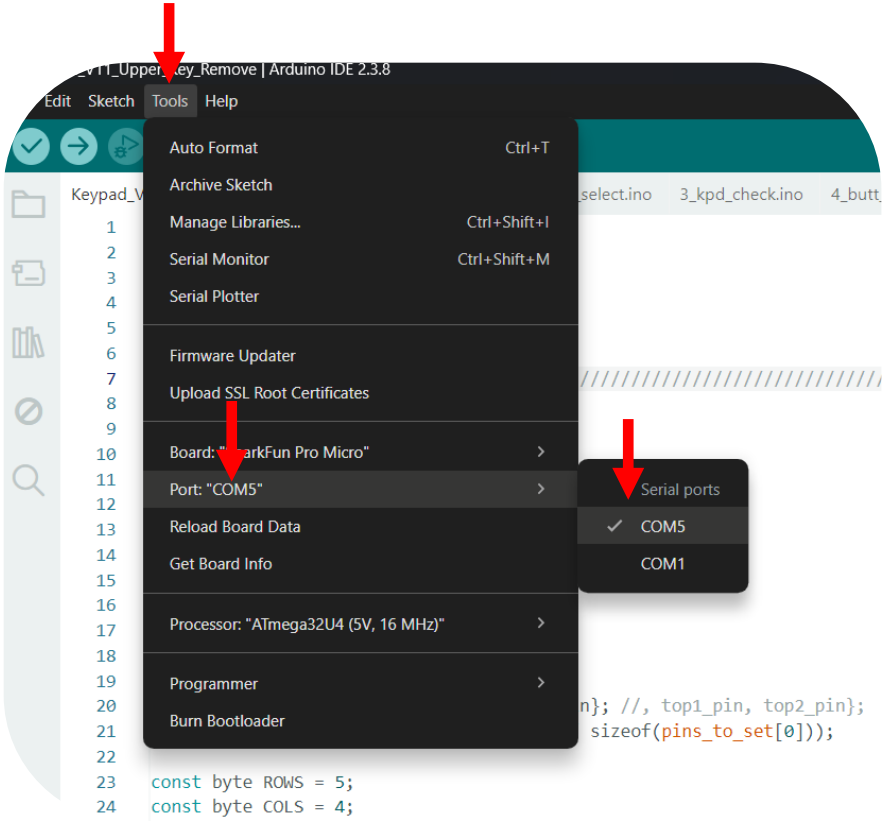




Connecting The Board

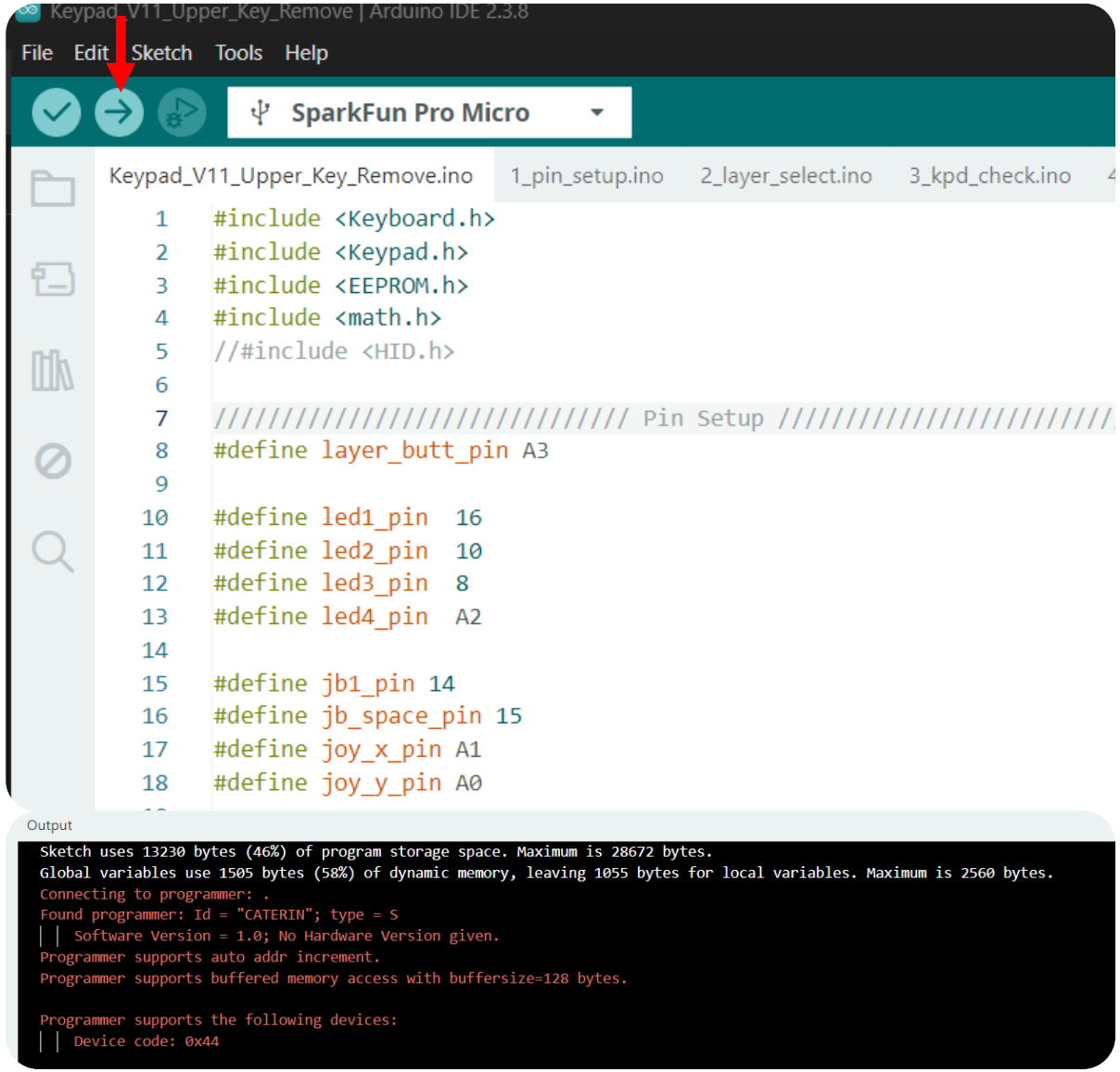
- Plug the Arduino into your computer using a micro-USB that can transfer both power and data
- Select the Arduino port: Tools -> Port -> COM”X”
  - Your COM port number may be different.
  - If you have more than one COM port, unplug the Arduino and see which one disappears. That’s the one for your Arduino.





Upload the Sketch to the Arduino

- With the Arduino connected upload the sketch.

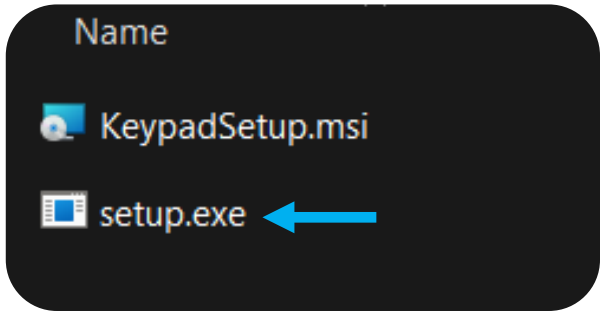


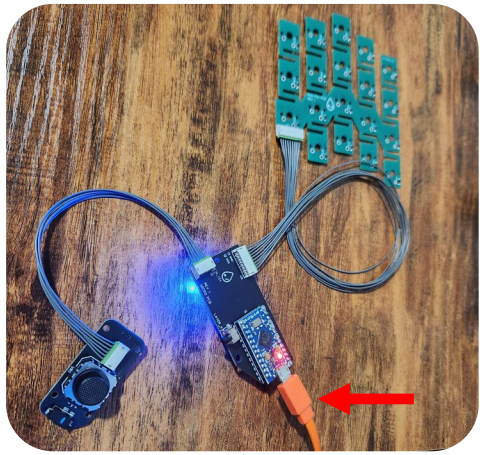
6

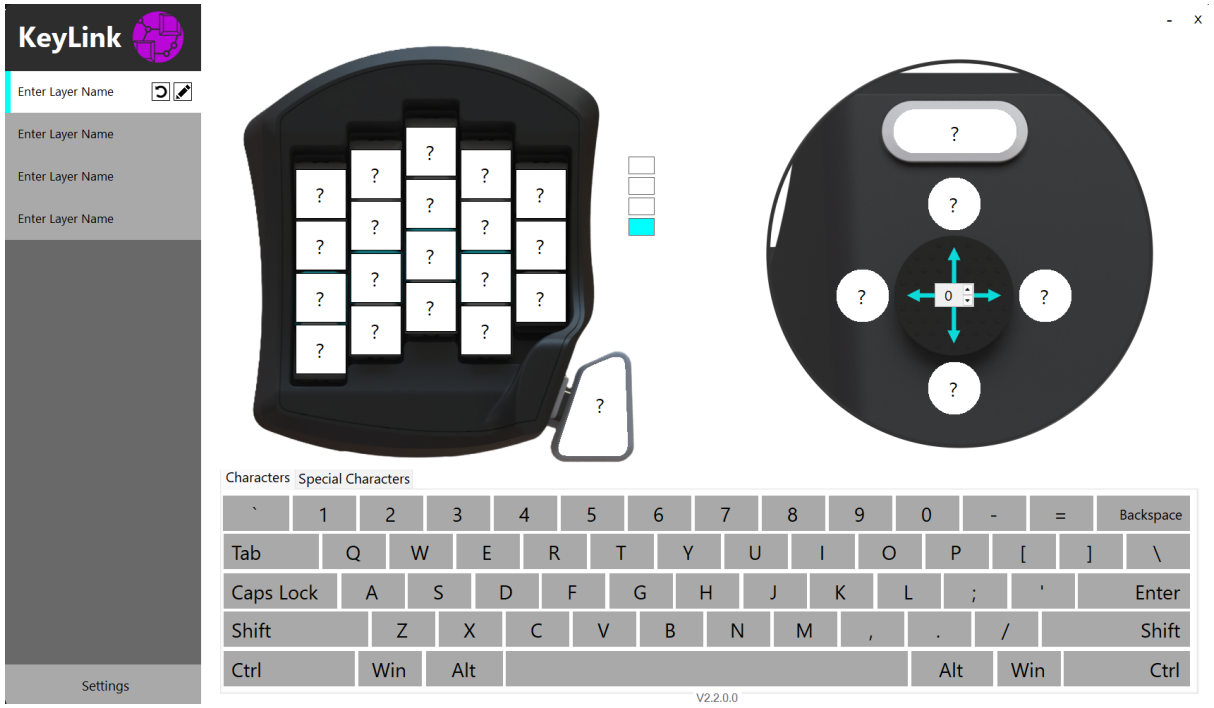
PC Software Setup

Install KeyLink

- Runk the setup file to install Keylink
  - Follow the setup instructions on screen
- Make sure your Arduino is still connected, then run Keylink







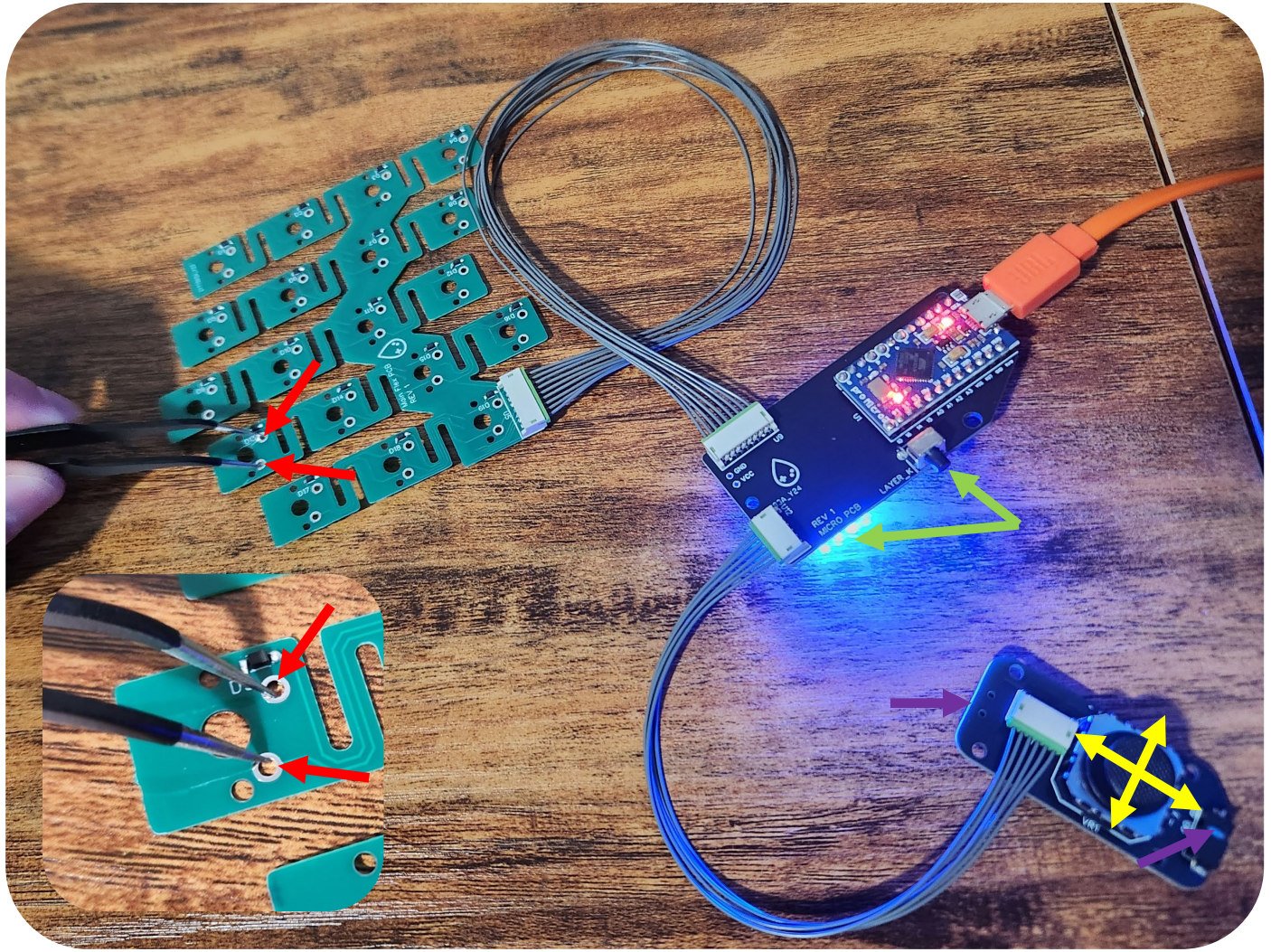
Program New Device

- Select Settings -> Program New Device -> Yes
  - The question marks should populate with characters



Test Programing of Device

- Open a blank work or notepad program to test typing with the keypad
- Test each of the 19 keys on the main PCB by electrically connecting the two plated through holes.
  - This can be done with any conductive metal like tweezers.
  - Note: modifier keys such as shift and escape do not produce a visible character
- Slide the joystick in each of the 4 directions
- Press the thumb cluster buttons on each side
- Press the layer button and check that the LEDs transition from 1 to 4, indicating the 4 stored layers.
- Repeat they key checks on the 4th layer (assuming you did the check already on the 1st).
- If any buttons don't activate, check your solder joints or that the diodes are soldered in the correct orientation
- Disconnect the cables from the Micro PCB

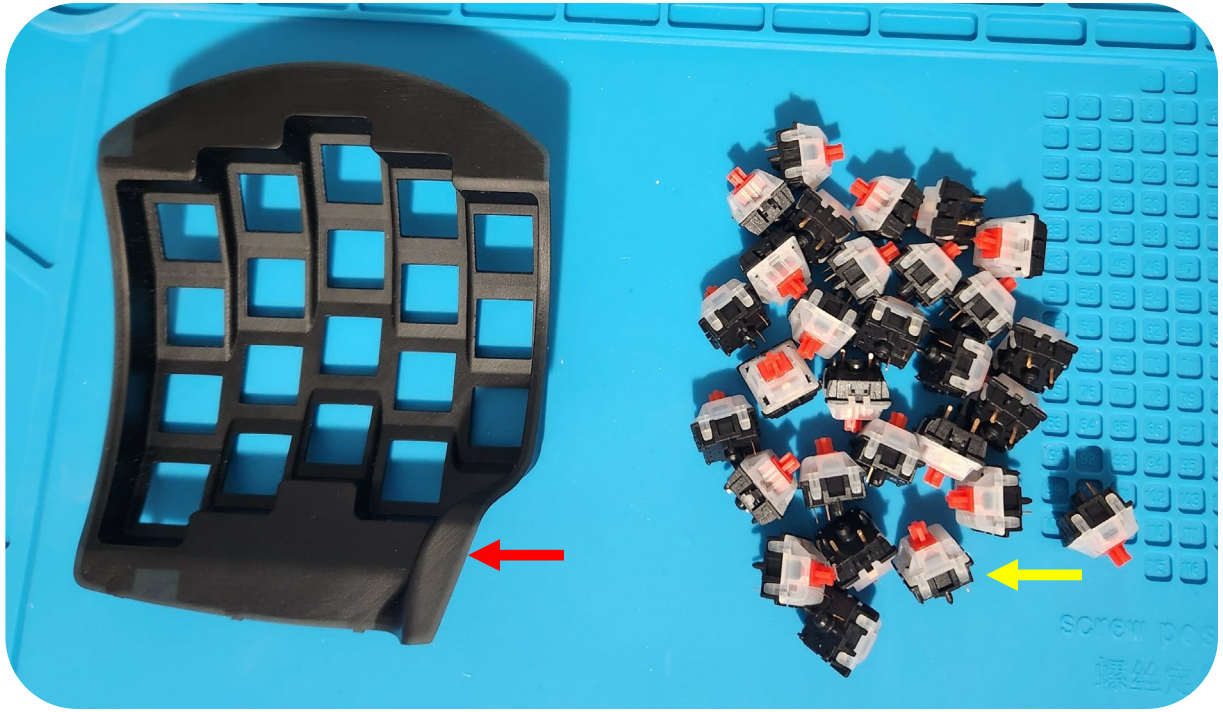


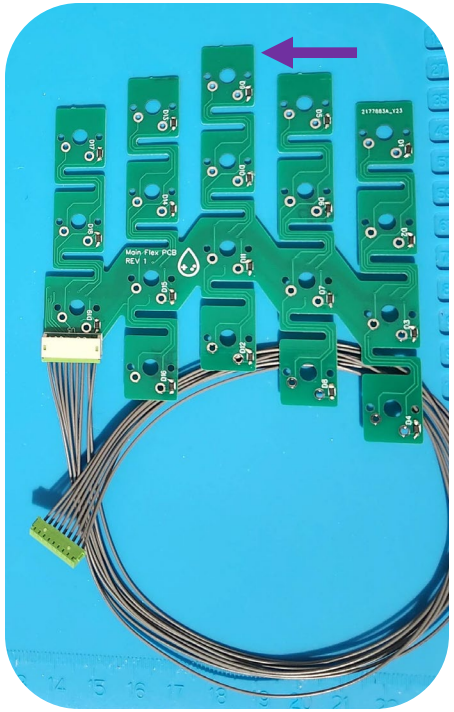
7

Key Mount Housing Sub Assembly

Components

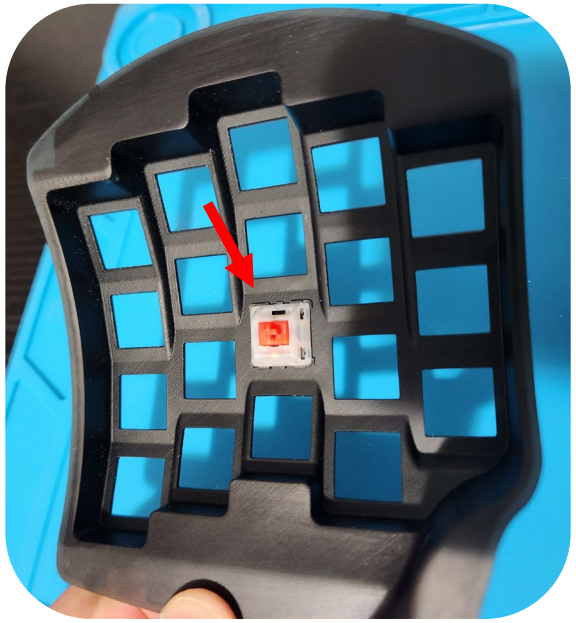
- Key Mount Housing
- 19X Mechanical Switches
- Main Flex PCBS

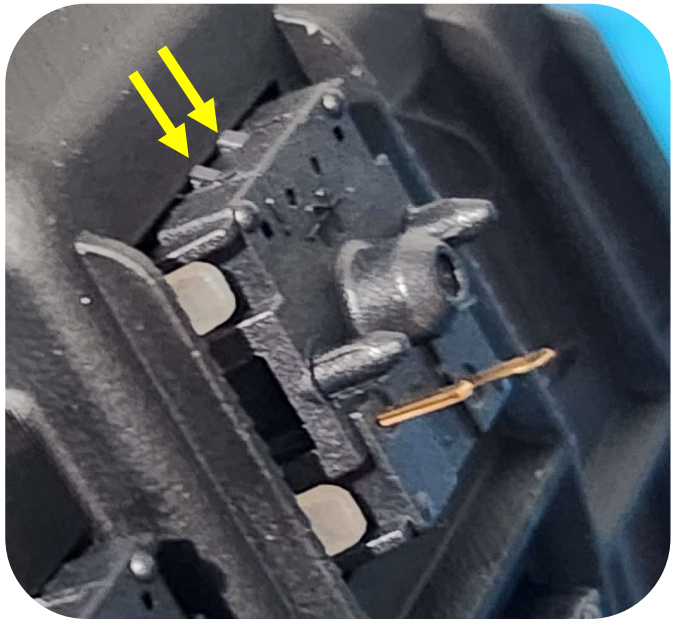




Switch Attachment to Housing

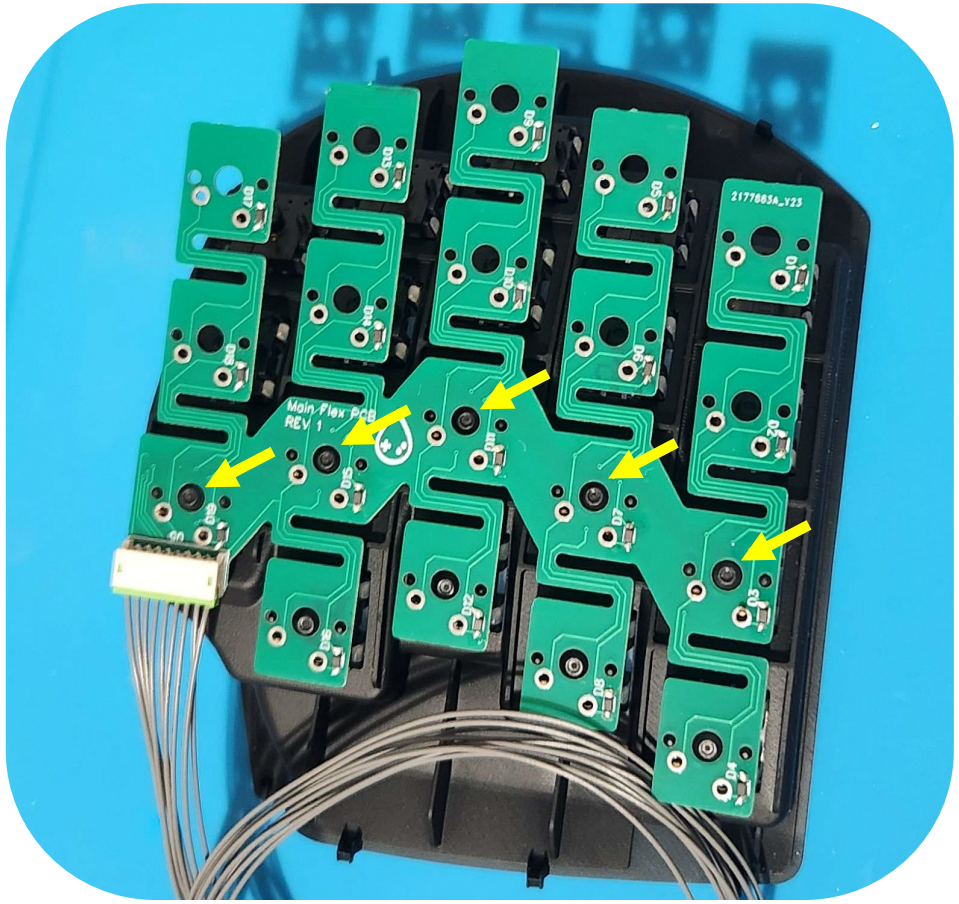
- Push the mechanical switches into the housing
- Flip over to ensure the contacts are facing the indicated direction. Take care not to bend the contacts
- Ensure the switches are fully seated buy checking that the snap fits on the front and back face have fully engaged the housing

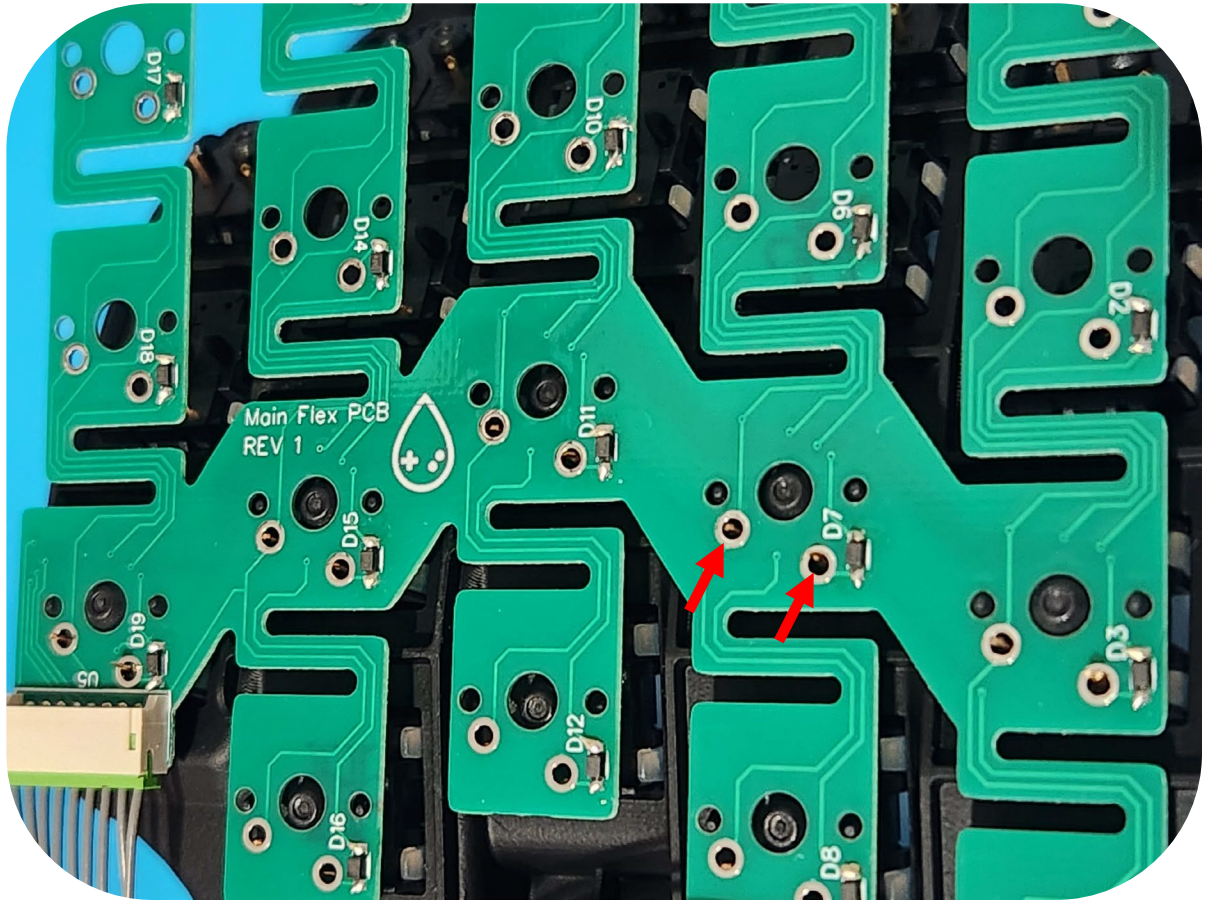





PCB Attachment

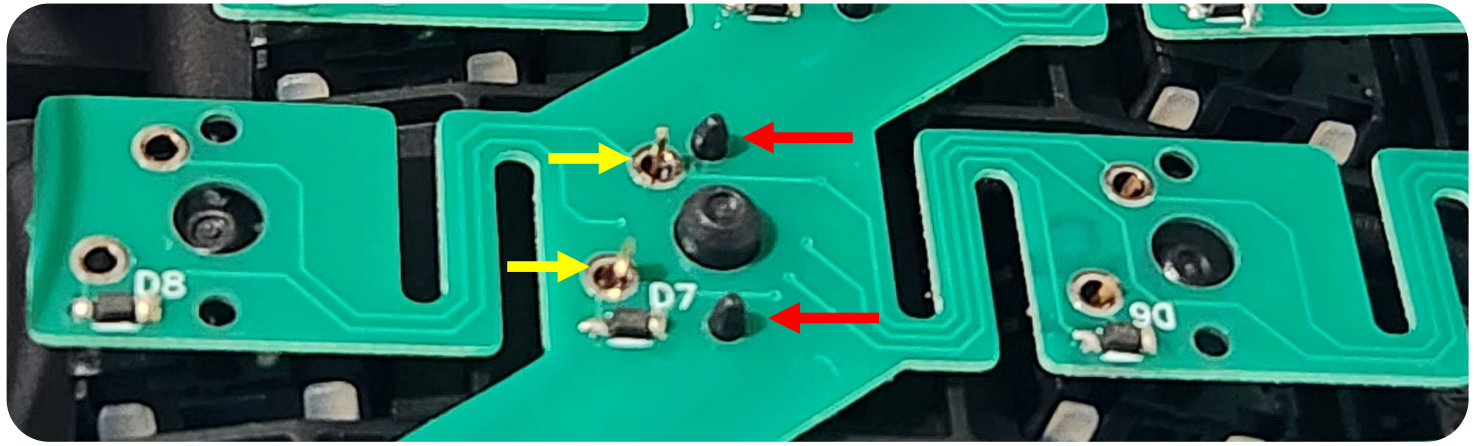
- Place the Main Flex PCB onto the housing as indicated below. It should rest aligned on the tallest row of switch pegs
- Ensure that the switch pins are aligned with their respective PCB through holes. Bend the pins to align if needed
  - The lower rows of pins will appear slightly miss aligned because they are on a different plane

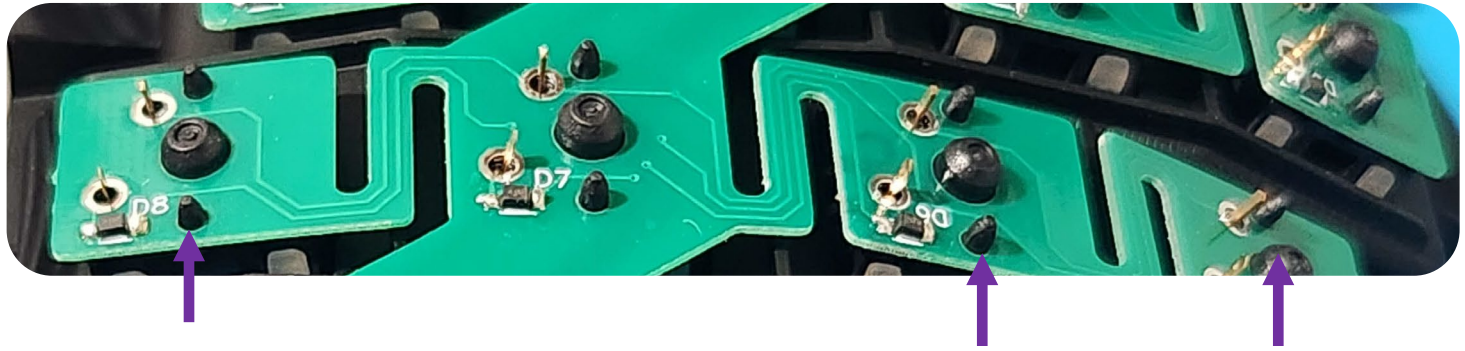




PCB Attachment Continued

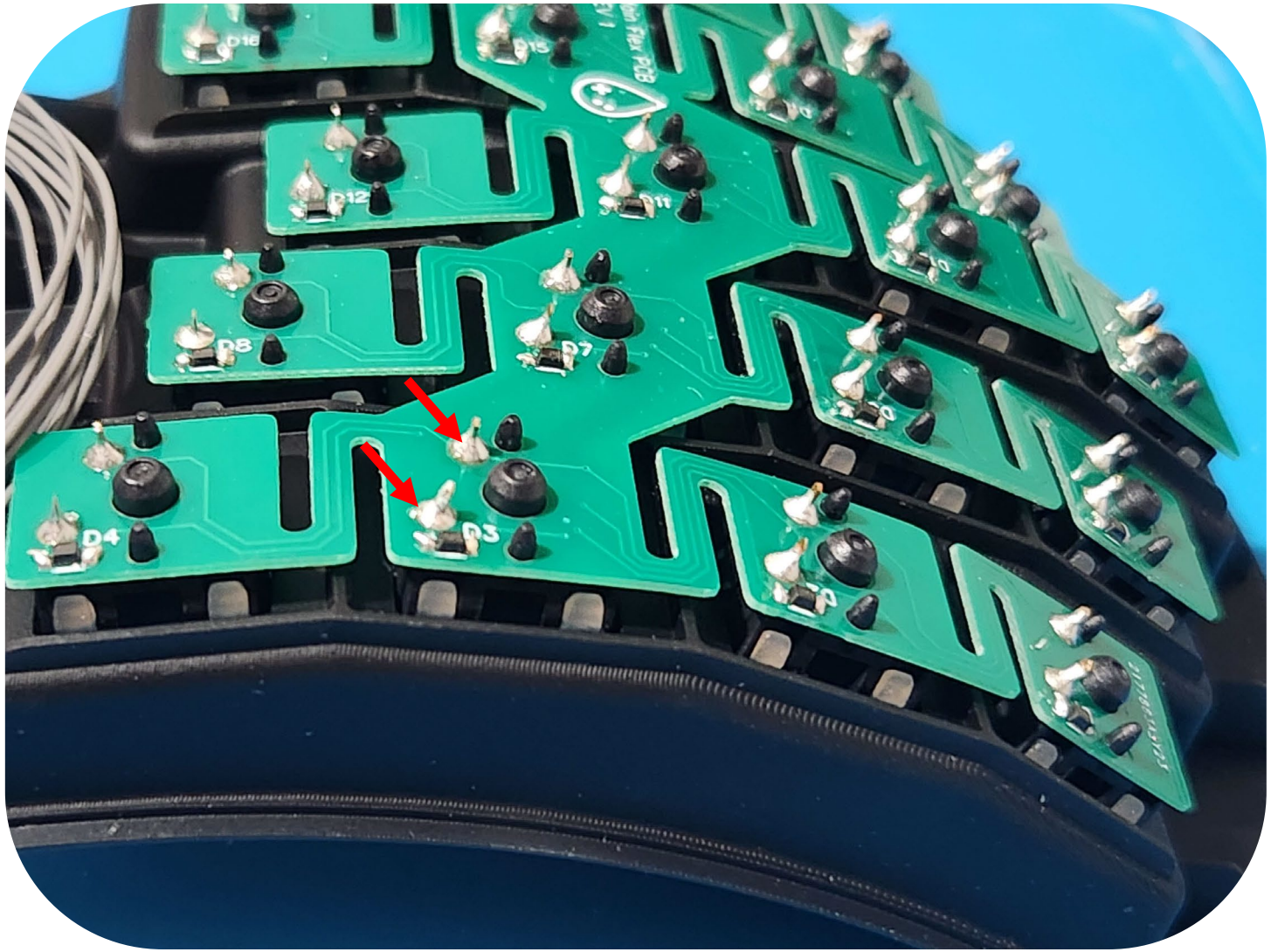
- Support the mechanical switch from the underside and press the PCB onto its pegs until they bottom out.
- Ensure the pins are poking through the PCB through holes
- Repeat for all rows and columns





Electrical Connection

- Solder each of the pins to the Main Flex PCB



8

Outer Main Housing Sub Assembly

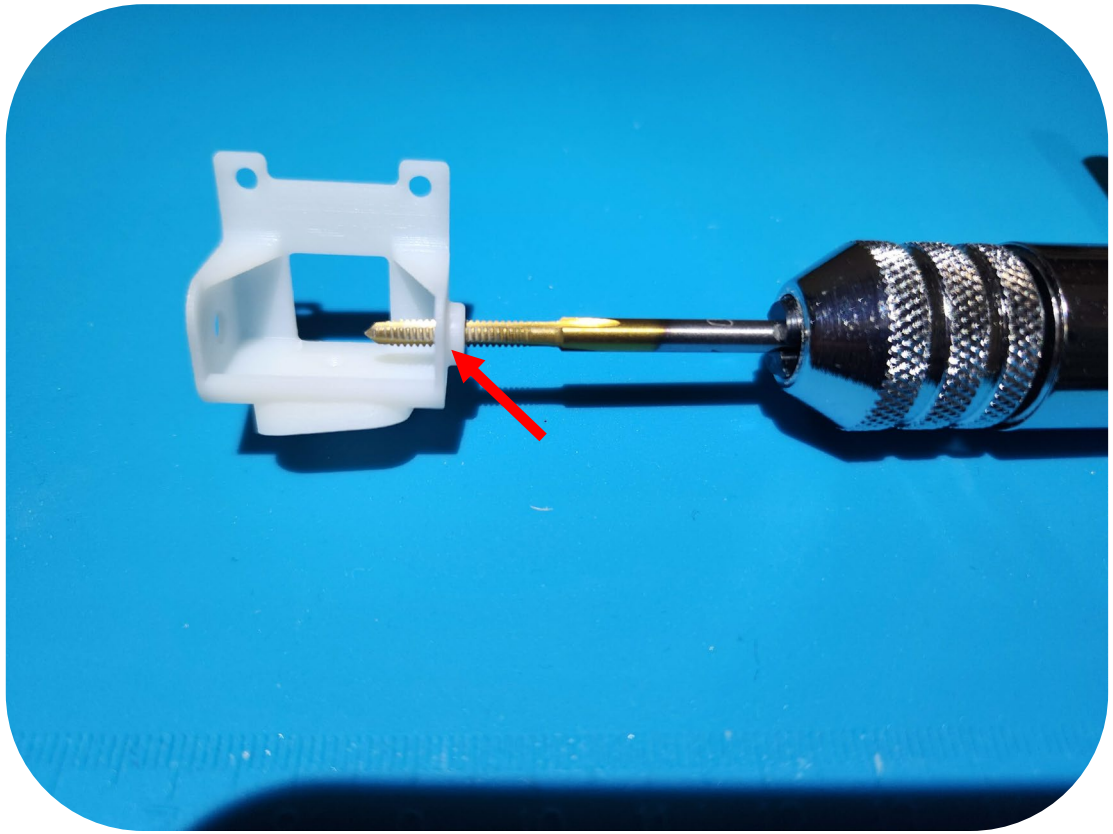
Components

- Outer Main Housing
- Thumb Cluster PCBA
- Thumb Bracket
- Thumb Switch
- Thumb Button
- #4-40 X 7/8 Sholder Screw
- 4X #2 X ¼ Plastite Screw



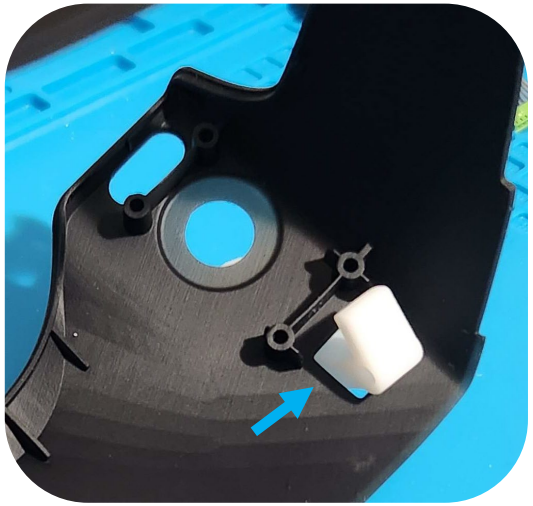
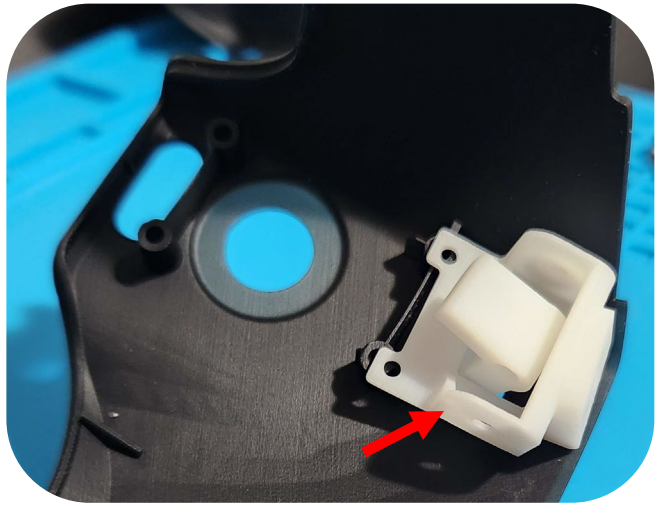
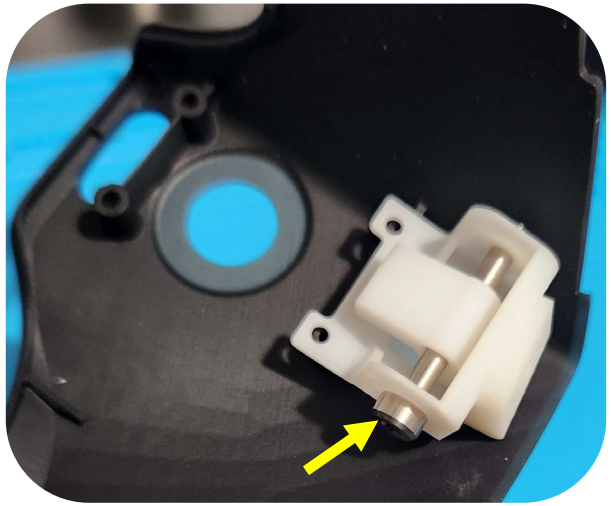
Tap Bracket

- Using a 4-40 Tap, thread the indicated hole on the thumb bracket



Thumb Interface

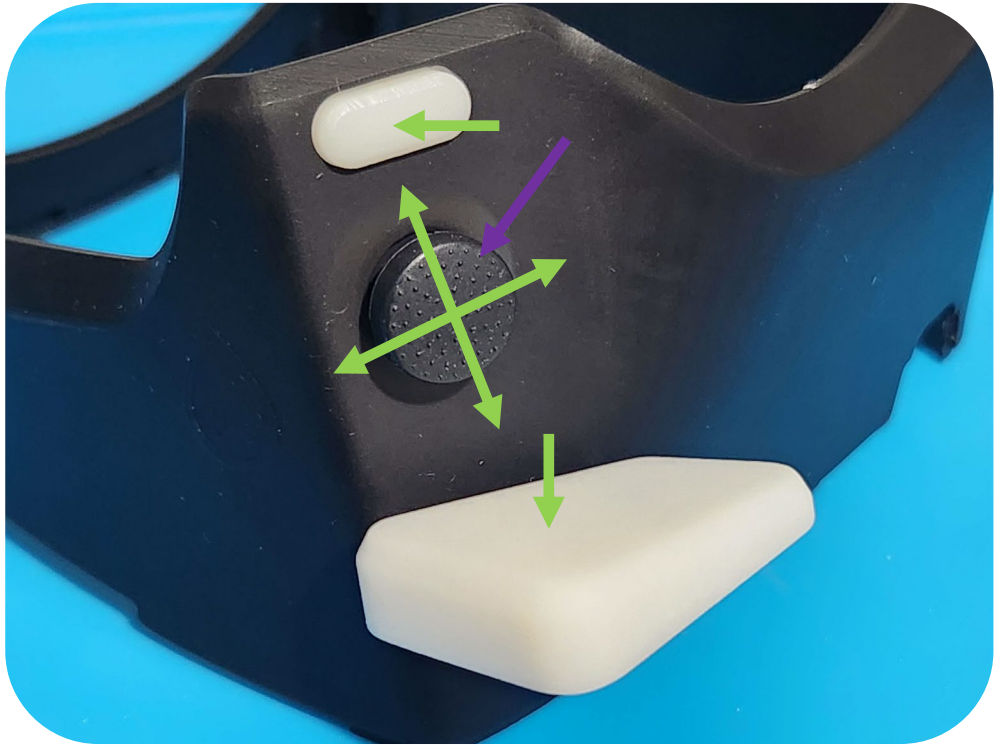
- Place the thumb switch through the hole shown in the housing
- Slip on the thumb bracket
- Attach the two parts with the shoulder screw
- Set the thumb button.
  - Note the notch orientation at the top


PCB Attachment

- Place Thumb PCBA onto the housing to sandwich all components
- Ensure the thumb switch arm is resting against the blue PCBA button head
- Secure with 4 screws
- (Optional) place 3rd party cap of joystick head
- Test that the joystick moves freely and the two white buttons actuate (click)



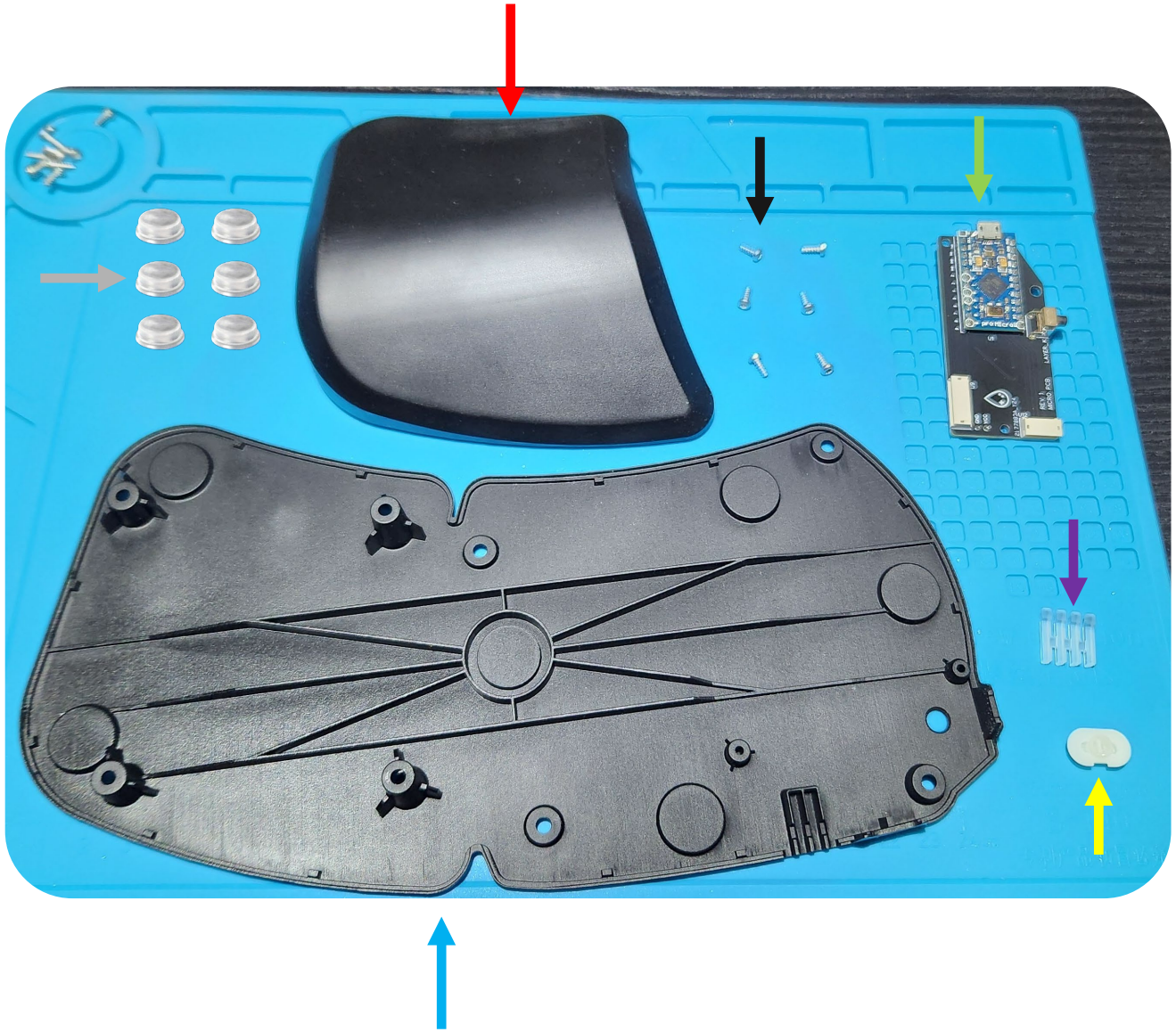


9

Housing Base Sub Assembly

Components

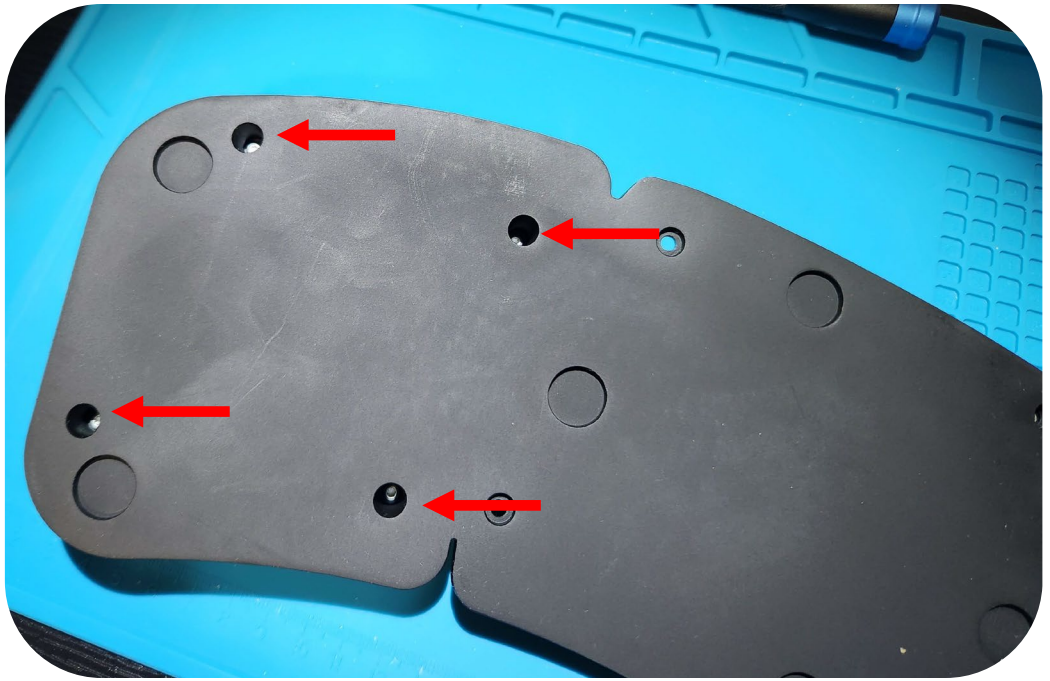
- Housing Base
- Wrist Rest
- Layer Button
- Layer Light Pipe
- Micro PCBA
- Adhesive Backed Rubber Bumpers



Wrist Rest

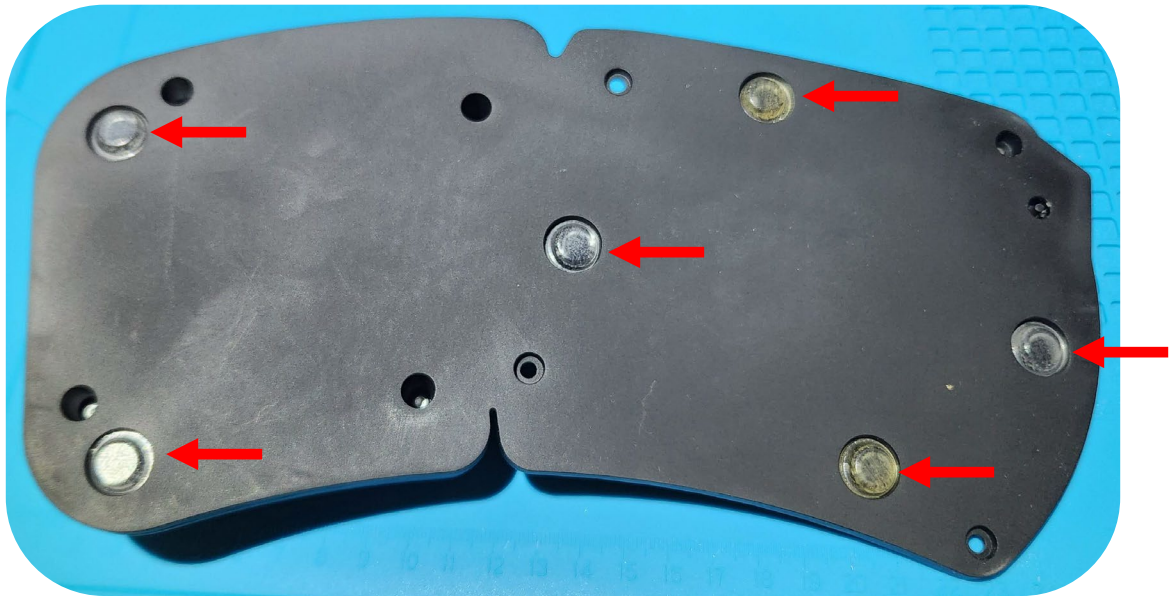
- Place the wrist rest onto the housing base
- Secure with 4 Screws





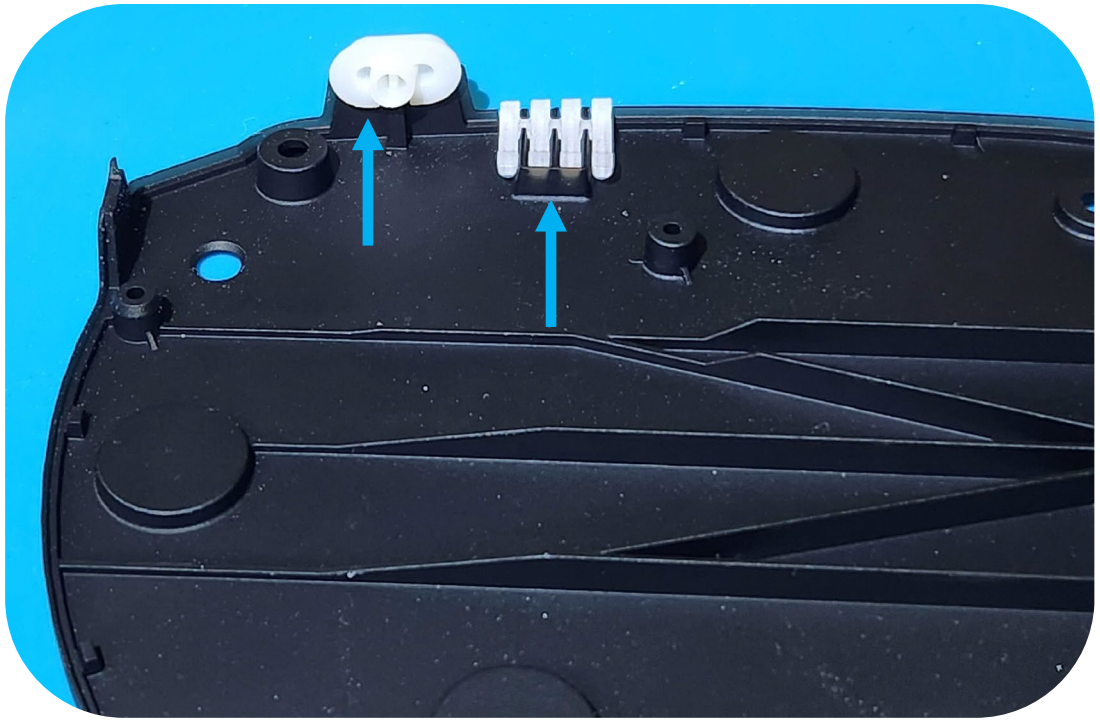
Feet

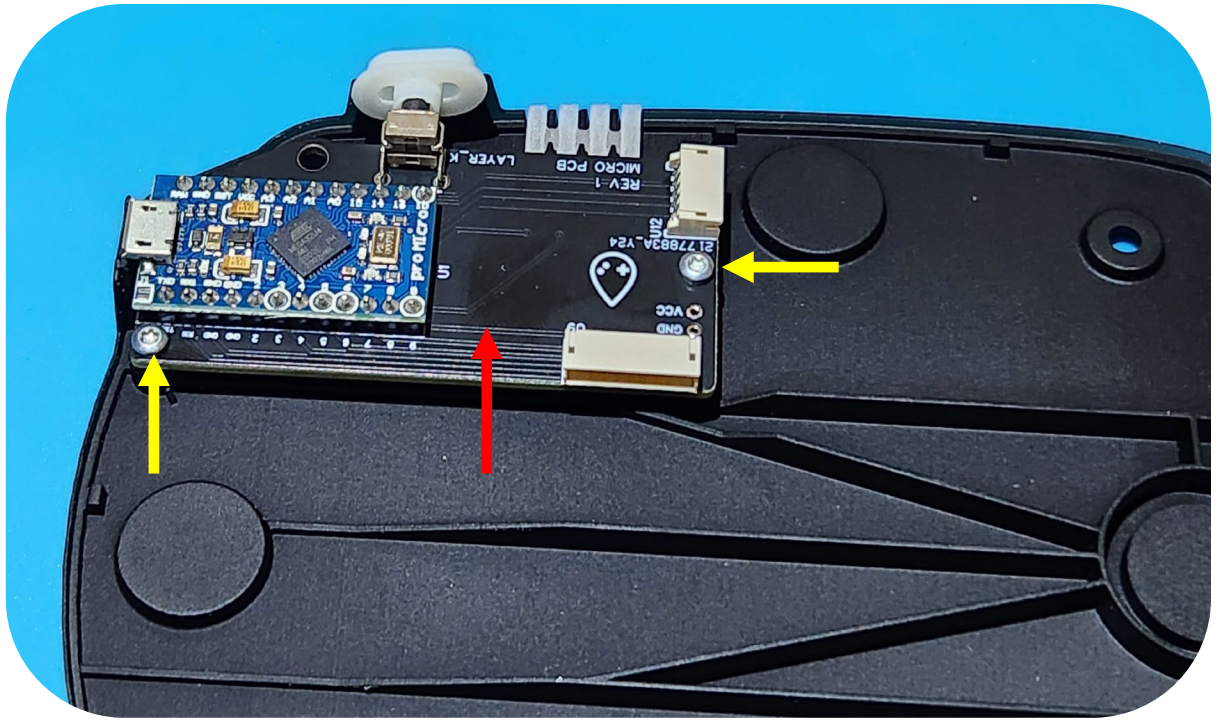
- Adhere the adhesive backed rubber bumpers into the indicated recesses



Micro PCBA Attachment

- Place the layer button and layer light pipe into the housing base
- Place the Micro PCBA into the housing as indicated
- Secure with 2 screws





10

Top Level Assembly

Components

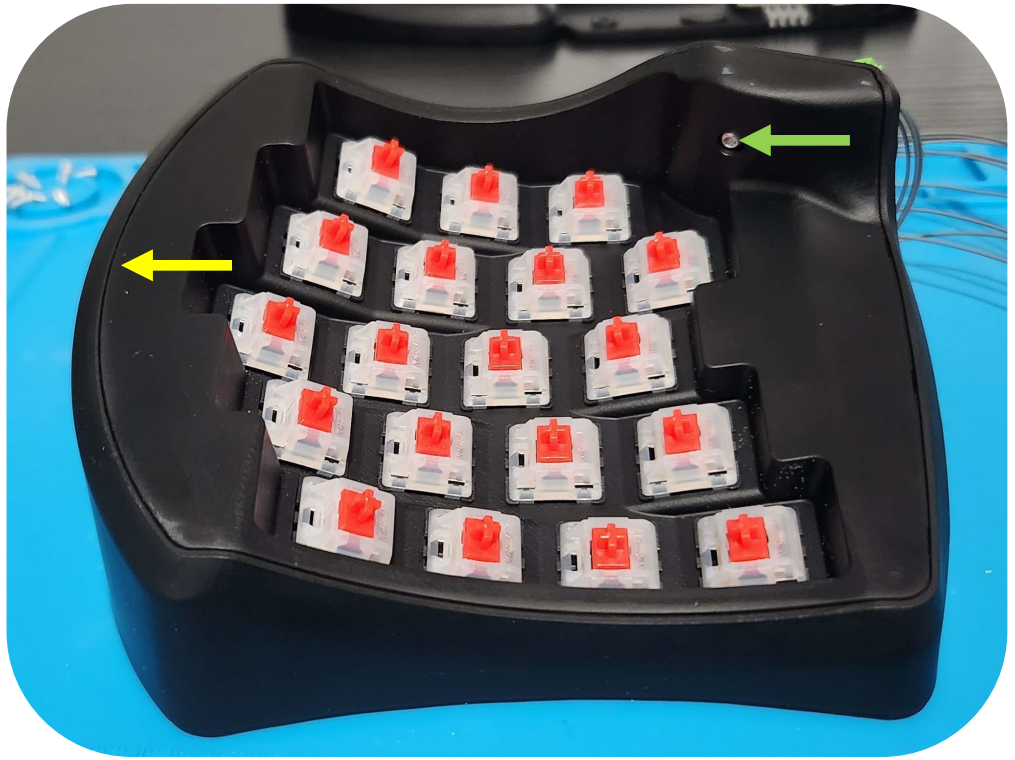
- Housing Base Sub Assembly
- Key Mount Housing Sub Assembly
- Outer Main Housing Sub Assembly
- Keycaps
- 5X #2 X ¼ Plastite Screw



Key Housing Assembly

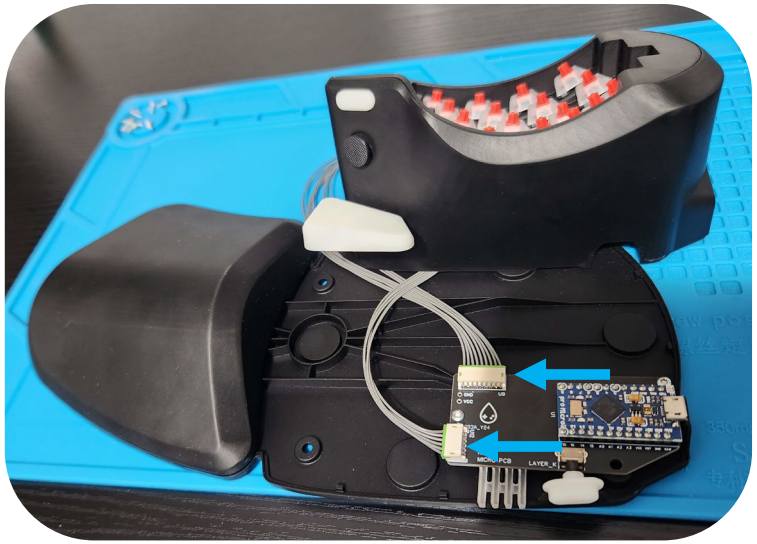
- Toe the Key Mount Housing Sub Assembly into the back of the Outer Main Housing Sub Assembly as shown
- Press down on the front end to engage the snap fits between the two housings
- Secure with a screw



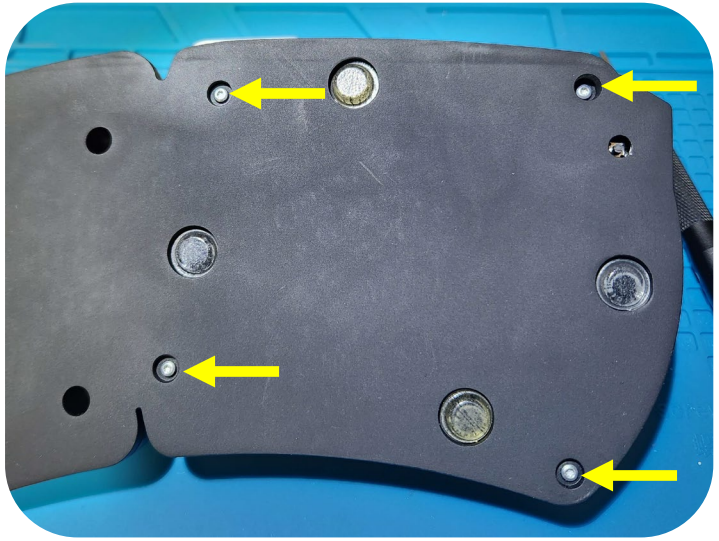


Main Housing Assembly

- Connect the two cable to the Micro PCBA
- Align the housings ensuring the layer button can still move
- Secure with 4 Screws







Key Cap Assembly

- Pick your favorite keycaps and assemble onto the switches
  - It is recommended to use all Charry R4 row keycaps





Happy Gaming





11

KeyLink Tutorial



KeyLink UI

- Select the layer you wish to edit. There are 4 layers stored on board the keypad
- To reprogram a key, click on the key you wish to change and select the character you wish to change it to on the virtual keyboard. Special characters are on a separate tab
- To change the angle of the joystick buttons, use the arrows at the center of the joystick icon. Currently the joystick acts as a D-pad
- Resets the keys to their default settings
- Renames the selected layer
- Resets all layers to default settings

